The Reserve Bank of Australia is expected to hike rates by 25bp to 4.85%. Elsewhere, markets will be watching a raft of key regional data releases, including China's PMI figures, South Korea's trade and inflation data, Japan's wage growth numbers and Taiwan's latest inflation reading

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RBA set to hike as inflation proves more persistent

Australia: RBA expected to hike rates by 25bp

We expect the Reserve Bank of Australia to raise its policy rate by 25bp to 4.85% on Tuesday rather than wait until December. Headline inflation rose by 90bp between July and September to 4.3% year-on-year. Fuel prices largely drove the increase, while goods, services, and core inflation also picked up. With underlying inflation not yet on a sustained downward path, the case for a pause remains limited.

China: Official PMIs expected to soften slightly

China releases October purchasing managers' index data on S… P… R… o… re…

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South Korea releases October trade data over the weekend. Market consensus expects export growth to accelerate to 89.4% YoY, supported by continued strength in semiconductor and AI-related demand. Imports are expected to rise to 41.1%. The trade surplus is expected to narrow to US$38.3bn from a revised US$50.0bn.

October inflation data will be released Tuesday. Market consensus expects headline CPI inflation to ease slightly to 2.8% YoY from 2.9%, while core inflation slows to 2.7% from 2.8%. Inflation would nevertheless remain above the Bank of Korea's target. Strong

exports also should continue to support growth and the case for maintaining a restrictive policy stance.

Japan: Wage growth expected to remain firm

Japan releases its September labour cash earnings data on Thursday. Market consensus expects nominal wage growth to ease slightly to 3.8% YoY from a previously revised 3.9%. Real earnings growth is seen slowing to 1.3% from a revised 1.6%. Firm wage growth should support further Bank of Japan tightening amid elevated inflation and negative real interest rates.

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Country	Time (GMT+8)	Data/event	ING	
		Saturday 31 October		
China	0930	Oct Manufacturing PMI	49.8	50.1
	0930	Oct Non-manufacturing PMI	50.1	50.2
		Sunday 1 November		
South Korea	0800	Oct Exports (YoY%)	CF: 89.4	83.5
	0800	Oct Imports (YoY%)	CF: 41.1	26
	0800	Oct Trade Balance (USD bn)	CF: 38.3	50
		Monday 2 November		
Australia	0830	Sep Building Permits Prelim (MoM%)	-	-6,1
China	0945	Oct RatingDog Manufacturing PMI Final	CF: 52	52,1
India	1300	Oct IHS S&P Global Manufacturing PMI Final	-	54,9
Malaysia	0830	Oct IHS S&P Global Manufacturing PMI	-	49,9
Indonesia	0830	Oct IHS S&P Global PMI	-	52,4
	1200	Oct CPI (MoM%/YoY%)	0.3/3.3	0.3/3.3
	1200	Oct Core CPI (YoY%)	2,7	2,8
	1200	Sep Trade Balance (USD bn)	-	3,6
	1200	Sep Exports Growth (YoY%)	-	6,7
	1200	Sep Imports Growth (YoY%)	-	19,1
Philippines	0830	Oct S&P Global Manufacturing PMI	-	49,6
Singapore	2100	Oct Manufacturing PMI	-	51,7
Taiwan	0830	Oct IHS S&P Global Manufacturing PMI	-	56,7
Thailand	0830	Oct S&P Global Manufacturing PMI	-	54,3
South Korea	0830	Oct IHS S&P Global Manufacturing PMI	-	53,9

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Taiwan Korea Japan China Austria

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Upside risks to inflation and downside risks to growth clearly complicate the ECB's job. However, comments at today's press conference suggest the ECB is set to hike again in December

Carsten Brzeski

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nuanced: despite growing stagflationary pressures, the ECB still looks set to hike rates again in December.

The decision to keep interest rates unchanged was not a surprise. The main argument behind this decision is that there are currently few knock-on effects from higher headline inflation, while financing conditions, mainly bond yields, have become more restrictive.

A pause after two rate hikes this year was expected, even if macro developments since the last meeting could have justified another rate hike. Headline inflation increased again in September, and

most leading indicators point to continued resilience in the eurozone economy. However, the surge in bond yields on the back of the French crisis is already imposing a new restrictive monetary policy stance. Or in other words, the bond market has taken over the ECB's job.

According to Christine Lagarde, risks to the eurozone's growth outlook remain tilted to the downside, while risks to the inflation outlook are tilted to the upside. A combination of risks that clearly complicates the ECB's path ahead. This dilemma was underscored by her observation that key developments since the September meeting have been a further rise in energy prices and a worsening o

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1. The length and intensity of the war in the Middle East. The longer oil prices remain elevated, the higher the probability of more rate hikes.
2. The knock-on effects of higher energy prices. So far, there has hardly been any pass-through of higher energy prices in the rest of the economy. If this remains the case, the need for further rate hikes drops. If it doesn't, the possibility of rate hikes rises.
3. France. Any solution to France's current fiscal woes could bring some relief to bond markets and allow the ECB to continue hiking rates. Without any solution, bond markets have taken

over the ECB's role, making rate hikes less likely (and less needed).

With today's comments, another rate hike at the ECB's December meeting is clearly in the offing. It would require a sharp worsening of the growth outlook to probably prevent the ECB from hiking once again.

In short, with downside risks to the growth outlook and upside risks to the inflation outlook, the ECB seems to be increasingly concerned about stagflationary forces, without calling them stagflationary. Remember back in April, Lagarde called 'stagflation' a

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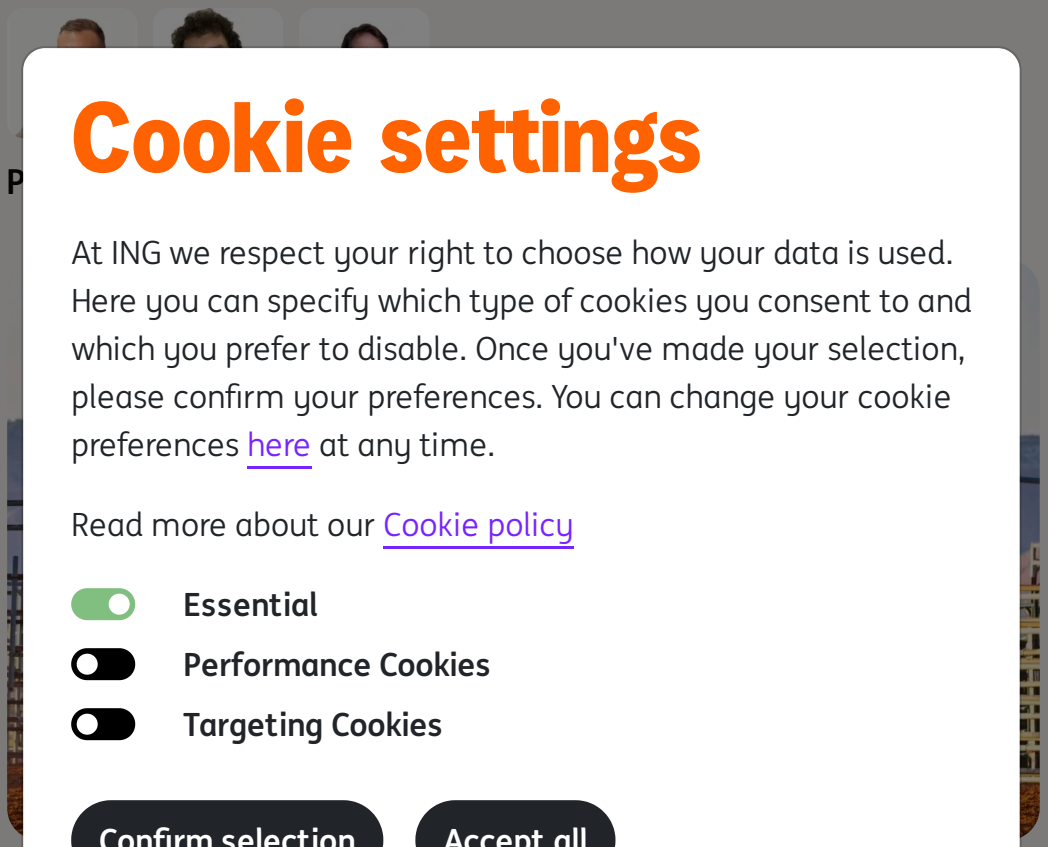
Could the US 10-year Treasury hit 6% in 2027? And could markets be too relaxed about euro rates? AI-driven growth, stubborn inflation and extraordinary government borrowing could keep global rates under pressure, while France's debt challenge adds another potential fault line in Europe. Join ING experts to explore what could drive rates in 2027



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The US 10-year Treasury hitting 6% in 2027 is not as far-fetched as it sounds. AI and productivity, extraordinary fiscal deficits, and stubborn inflation could combine to keep global rates under upward pressure. But which force will ultimately dominate? What could Treasury Secretary Bessent do if yields surge? And what if inflation proves stickier than market expectations suggest? Or is all that overblown, and we land in a less dramatic place? We explore the forces that could drive each scenario.

In the eurozone, the inflation shock may be changing rather than disappearing – is it, and if so, what are the implications? To what extent could second-round inflation pressures, heavy bond supply

and rising global real rates complicate the outlook, even if the ECB turns more dovish? Meanwhile, France's growing debt challenge adds another source of uncertainty for euro sovereign markets. How bad can this get for bond markets and rates? Will bond supply finally test investor appetite?

Join ING experts as we explore the key scenarios for US and euro rates in 2027, and where the biggest risks and opportunities lie. Sign up here.

Speakers

Padhraic Garvey, Regional Head of Research, Americas

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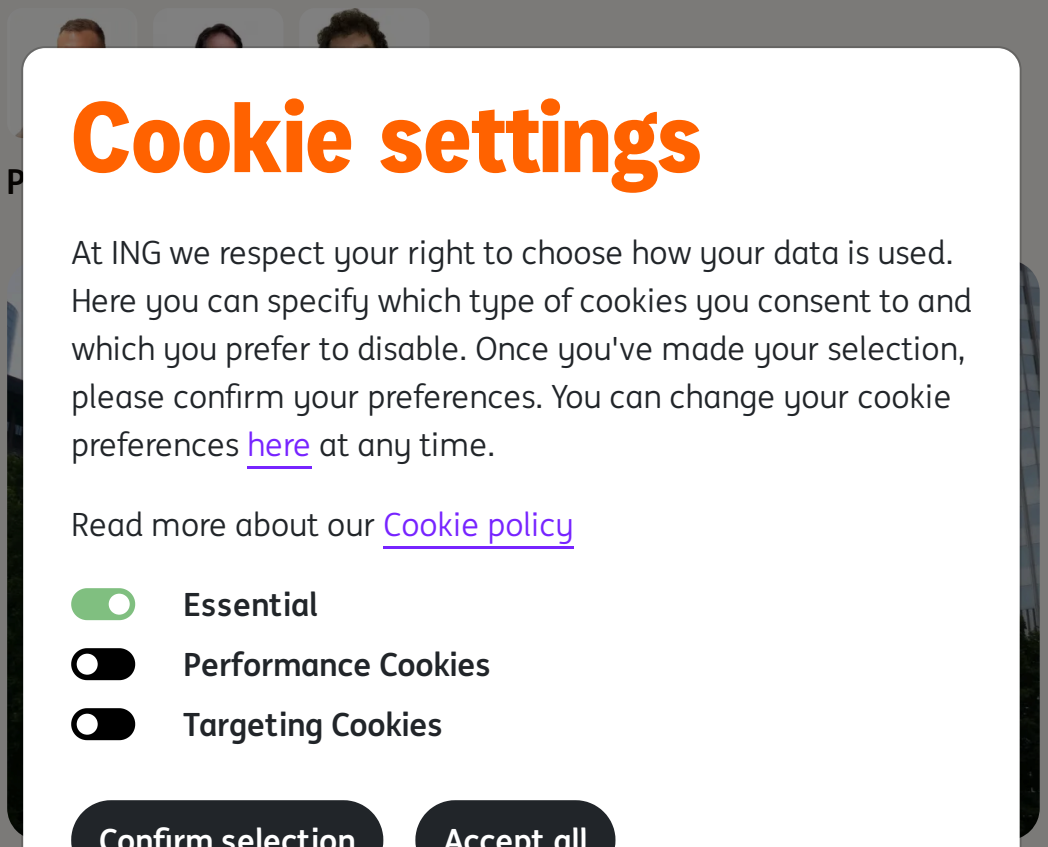
US Treasuries are in a mood to test the downside for yields after all. Still not sold on following through on it. But monitoring. The ECB meeting did not change our rates outlook and confirms a December hike is likely. We brace for more volatility as Friday's CPI and GDP readings from the eurozone are important ingredients in markets' rate hike expectations



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US Treasuries looking through the rate hike narrative for December

The US 10yr yield made a decisive gap away from 5%, but to the downside. This was not a fundamentally driven move, in the sense that Thursday's PCE inflation data continued to point to an ongoing significant deviation from where Chair Warsh would like to see it at (he reiterated in the aftermath of the FOMC meeting that he wants to see 2% on core, but it rose to 3.1% for September). The

income, spending and jobless claims data were also in line, and at the firm end of the range.

Instead, Treasuries are willing to remain quite responsive to the vagaries of the sentiment on AI valuations and equity market wobbles. Nothing terribly sinister went down here, but the ongoing mood music has just turned that bit sour. Meanwhile, the 7yr auction was fine, with no tail, and met decent demand.

What we can say is the move was more a back-end thing than a front-end one. Some chunky flows did go through on the buy side. We can't ignore this move, as it's contrary to our preferred morph b

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hawkish one. A Reuters report about ECB staff estimating that higher bond yields could cut cumulative growth by 100bp and inflation by 90bp in 2027-2028 relative to the September projection adds to the notion that the December hike could be the ECB's last of the cycle.

While this is also our baseline scenario, more needs to fall into place for markets to lean more in that direction. Easing energy prices and no evidence of second-round effects would certainly help, but especially on the latter, greater certainty needs to be established over months of tracking the data.

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Currently, the market still sees the upside inflation risks flagged by President Christine Lagarde as keeping the ECB on a hawkish footing. While the market's pricing for further ECB tightening over the next 12 months has dropped towards 65bp, this was mainly driven by French-related risk-off momentum and, of late, also growing AI jitters. As arguments for the ECB to change its tack, the spread widening thus far will likely not be sufficient, meaning the market will remain exposed to bearish risks.

Friday's events and market views

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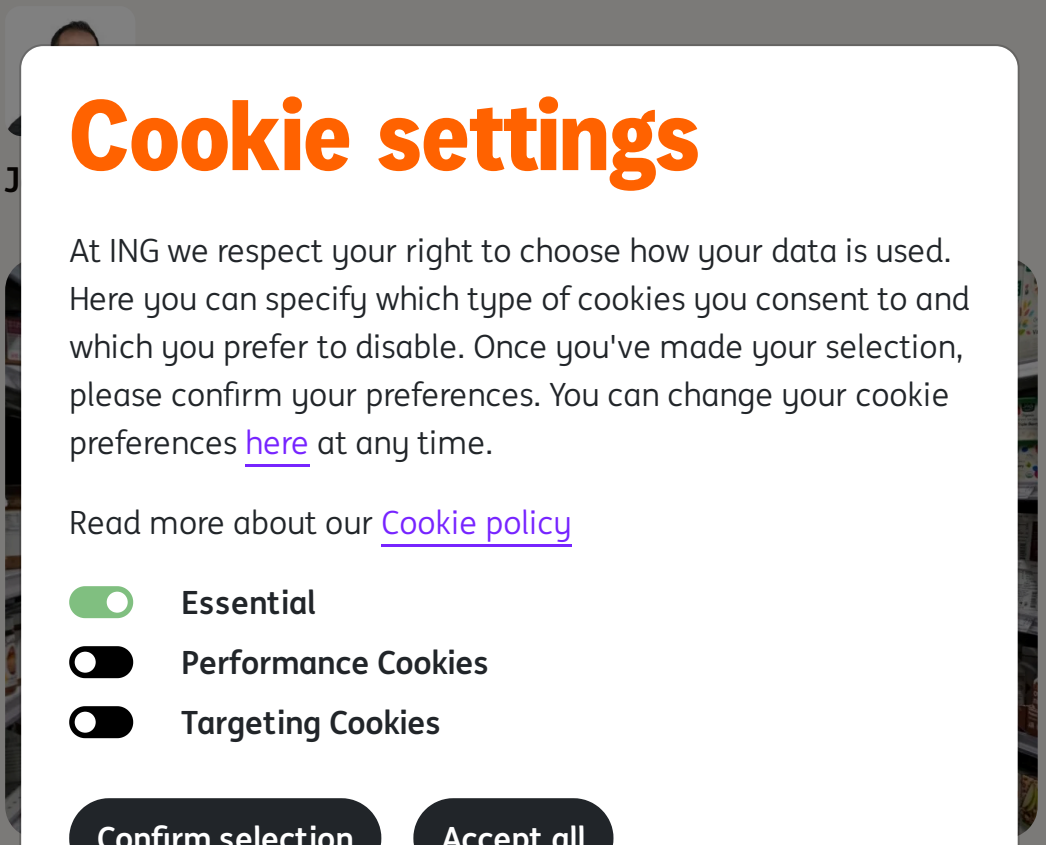
The US economy expanded robustly in 3Q, driven by strong consumer performance. Business investment surged, led by tech-related activity, but with many inputs imported, net trade fully offset the capex contribution. Cooler momentum points to slower 4Q GDP growth, but with inflation remaining hot, the Fed looks set to hike rates in December



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3.4% 3Q GDP growth

Hot consumer and investment spending held back by surging tech imports

Third-quarter US GDP growth was fractionally stronger than the market anticipated at 3.4% annualised (consensus 3.2%), while the

Fed's favoured inflation measure was 0.3%MoM/3.1%YoY in September, as expected.

Within the details, consumer spending reported a robust 3.8% QoQ annualised growth, but momentum was lost as the quarter progressed, based on the monthly profile, with September real spending flat on the month. Gross private investment was particularly strong, surging 19.4%, led by a 20.9% gain in equipment. This is overwhelmingly tied to the tech sector, as can be seen in the chart below.

Structures rose 5.4%, which is also tech related, being heavily o d 8 g r

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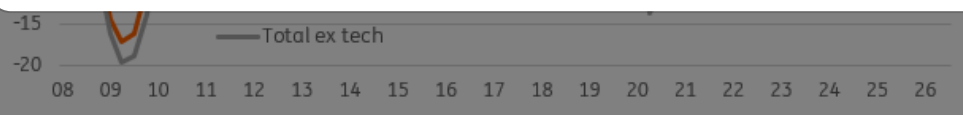

Source: Macrobond, ING

Inflation remains too hot for comfort

However, much of the high-value inputs for the tech roll-out are being imported from the likes of Korea and Taiwan. Goods imports leapt 26.4%, a massive drag on headline US growth. So, while gross private domestic investment contributed 3.3%pp to GDP growth, net trade fully subtracted 3.3pp, leaving a net flat outcome. This

narrative of tech investment being offset by tech imports looks set to remain a story in 4Q based on orders numbers.

The squeeze on spending power from weak wage growth and a rising cost of living meant real household disposable income fell 0.3%MoM in September, and this is likely to contribute to a cooler consumer spending story in 4Q, while government spending is also likely to partially reverse the 5.3% jump experienced in 3Q. As such, we are looking at 4Q GDP coming in closer to 2%.

On the inflation metrics, there was no surprise, with the CPI and PPI inputs clearly signalling the prospect of a 0.3%MoM increase. As the c… a… b…

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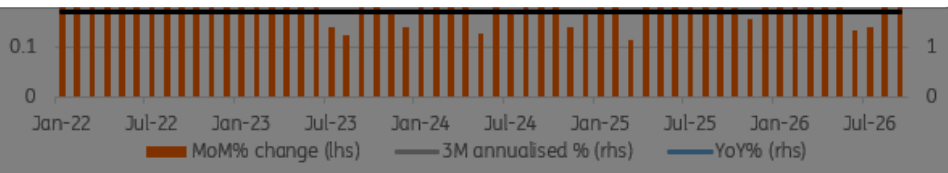

Source: Macrobond, ING

US | Inflation | GDP | Federal Reserve

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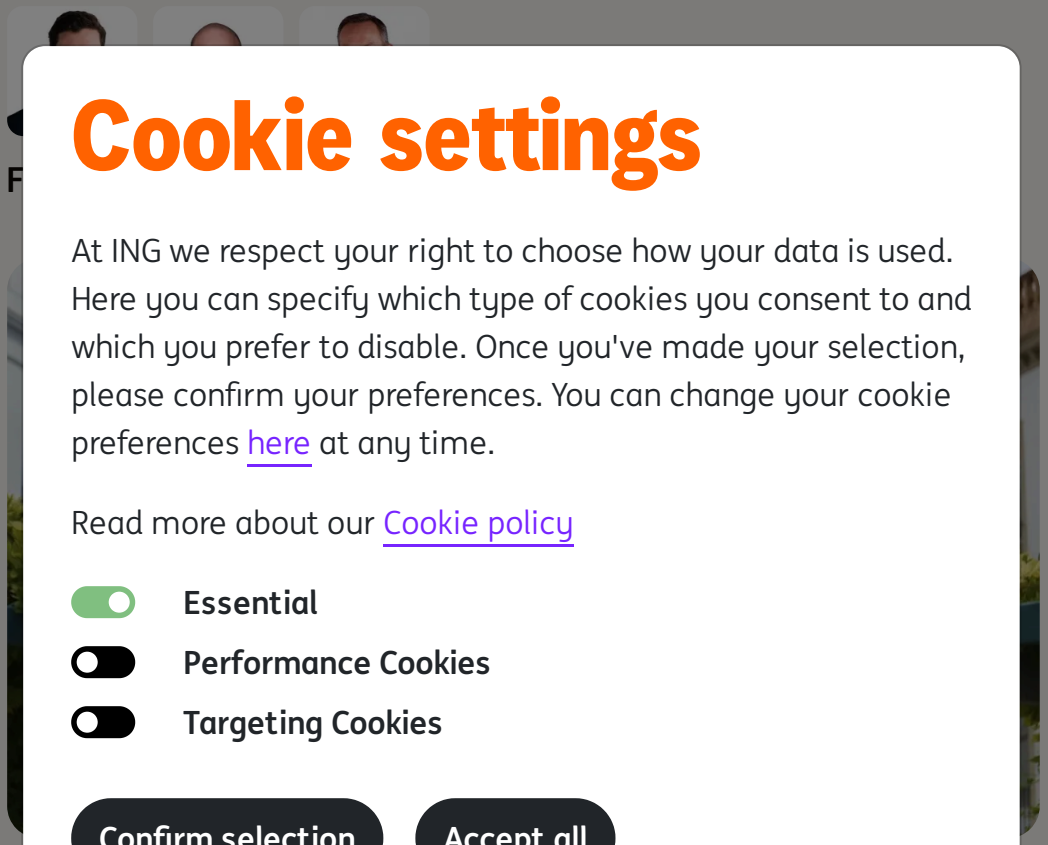
The ECB will likely keep rates on hold today, but we don't think French-related risk is enough to drive a material dovish shift yet, especially with energy prices running so high. We expect a contained euro impact. After a slightly negative reaction to yesterday's FOMC, we see the dollar finding some support from today's US PCE and GDP data



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USD: Don't read too much into the softish reaction to the Fed

The dollar traded a little softer after the Fed's hold yesterday, with the USD 2yr swap rate falling around 8bp. As discussed in our review note, the overall message remained hawkish, with Chair Warsh reiterating his firm commitment to price stability, adding that current inflation "remains elevated" while the economy is "expanding at a strong pace". The modest dovish market reaction was probably due to the lack of clearer guidance towards a

December hike, although we've now learned not to expect that from Warsh.

On the vote split, Beth Hammack and Lisa Cook's dissents in favour of a hike were not particularly notable signals in themselves. Cook's vote is a bit of a surprise, however, given she was considered a more centrist member, and may fuel further ire from the Trump administration, which is still attempting to remove her.

From an FX perspective, the picture for the dollar doesn't change after this Fed meeting. A December hike remains in play (21bp priced in) and its prospect should prove fairly resilient to a potential o

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The European Central Bank is likely to keep rates on hold today. We've outlined four scenarios in our ECB cheat sheet: three involve a hold, while one very hawkish scenario sees a hike. Our baseline view is neutral. We don't think we are at the stage where French bond market turmoil can trigger a significant dovish turn by the ECB, especially with such elevated oil and gas prices (TTF hit €100/MWh in Europe yesterday).

We expect much of the market reaction to be driven by the ECB's tone on France and the prospect of another hike this year. Markets are pricing in 21bp for December, and in September officials signalled that a rate hike by year-end was in play in their

customary tip to the media a few hours after the rate announcement. Signalling that rates are “likely” to be raised in December would constitute a modest hawkish surprise in our view. However, we see a slightly less committal message as more likely, perhaps that a December hike is “possible” or “in play”.

Either way, we don’t expect this meeting to be a turning point for the euro. EUR/USD has shown it’s not yet ready to break below 1.110, and we don’t expect today’s ECB risk event to trigger a major leg lower. A slightly stronger USD on US data may, however, keep modest pressure on the pair.

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growth. Our base case therefore remains that rates peak at 2.75%.

For CAD, oil remains the key swing factor. High prices can keep December hike expectations alive and support the currency, but we remain bearish on oil and think a correction would trigger dovish BoC repricing. We therefore see limited room for CAD gains from the domestic story alone and retain our year-end USD/CAD target of 1.420.

Francesco Pesole

CEE: The Fed offers a chance to stabilise, but gas is inviting further weakness

The CNB's "blackout period" begins today in the Czech Republic, and the market is returning following yesterday's public holiday. On Tuesday, another board member, Jakub Seidler, confirmed his intention to vote for a rate hike at next week's meeting. Given that he has consistently voted in line with the board majority, we believe a 25bp rate hike – bringing the rate to 4.00% – is a done deal. In our CNB preview published yesterday, we noted that the market is primarily driven by energy prices, with the CNB having relatively little impact. As the Czech market resumes trading after

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European gas prices climbed above €100/MWh as mounting supply concerns rattle the market ahead of the 2026/27 winter

Warren Patterson

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following the US midterm elections next week. Reportedly, QatarEnergy has extended its force majeure for several buyers until the end of the year. It means significant LNG supply disruptions are creeping deeper into the heating season. In addition, several unplanned outages in Norway will add to supply concerns.

Furthermore, forecasts show colder-than-usual weather in Northwest Europe from next week, pushing heating demand higher. This will likely start leading to more sustained drawdowns in storage. EU gas storage is just shy of 75% full. While we've had a

couple of days of draws in October, winter storage draws have yet to fully get underway.

The key question is whether these higher prices will destroy demand. In power markets, we're already deep into the gas-to-coal switching range. Industrial demand also shows signs of softening. Chemical company Yara recently reported curtailments of ammonia production in Europe due to high gas prices. This led the company's European gas consumption to fall more than 7% YoY in 3Q26. This trend is expected across European industry, especially with gas prices even higher now.

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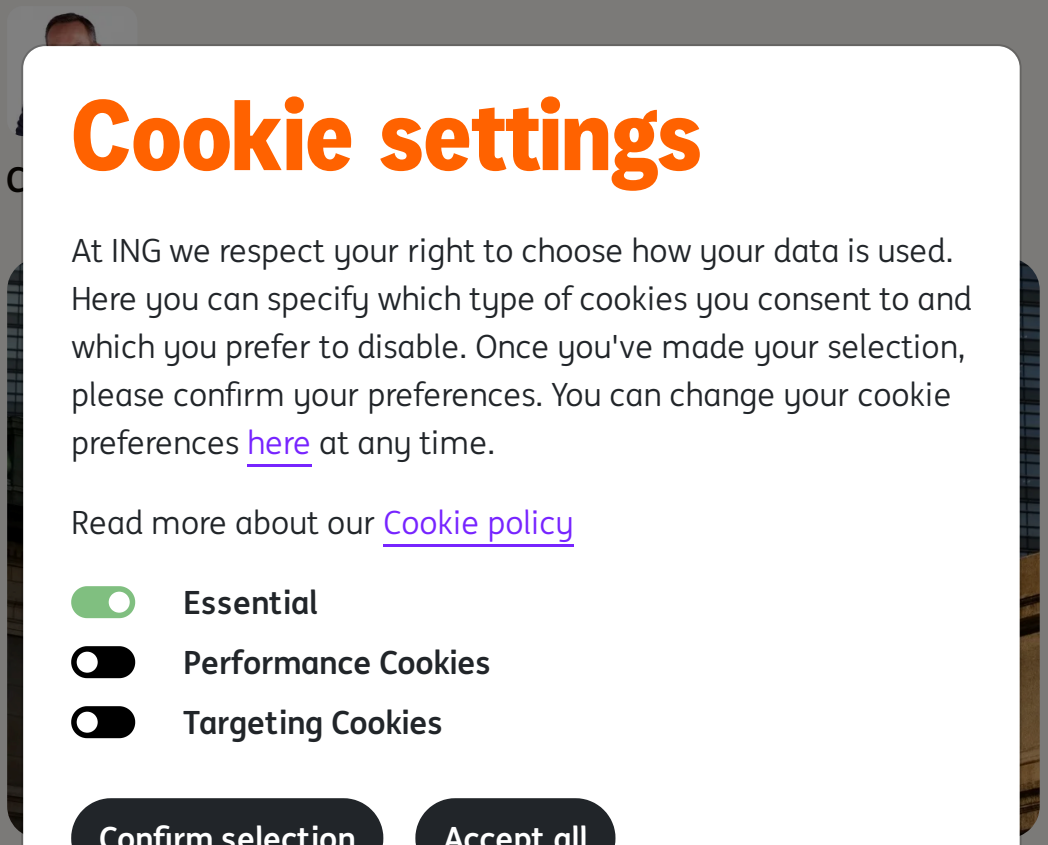
After September's policy rate hike to 1.25%, the Bank of Japan is expected to deliver a hawkish hold tomorrow. In focus will be the release of the October outlook report featuring the latest forecasts for growth and inflation. CPI projections above 2% for FY27 and FY28 should support expectations of rate hikes to 1.75-2.00% next year



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Price expectations well above 2%

Having hiked rates by 25bp to 1.25% in September, the central messages from the BoJ were largely similar; upside price risks remained, real interest rates were still negative and further policy rate hikes should be expected.

Perhaps the only reason that a rate hike is not expected at this 30 October meeting is that the BoJ has never delivered back-to-back rate hikes since it was given monetary policy independence in 1998. A hike in October could therefore smack of urgency.

Behind a widely expected unchanged decision will be a focus on the voting pattern. Did arch-hawk Hajime Takata again dissent in favour of back-to-back hikes? His vote for a hike would not be a surprise. Anyone joining him would.

More interest will be had in the October outlook report. The July report was pretty hawkish in that the median forecast for core (all items less food and energy) inflation for FY27 and FY28 were 2.6% and 2.2%, respectively. Let's see how they change in the new release.

Perhaps one comfort for the BoJ is that inflation expectations do n
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Market reaction

Markets currently price in an 80% chance of a 25b rate hike in December and then another 50bp by next September. While government representatives may well want the BoJ to go slow with its tightening, the case against the BoJ taking rates up to the 1.75-2.00% area is a weak one. Here the BoJ remains concerned with a weak yen, broadening AI demand and elevated energy prices.

There does not seem an obvious case for a big move in market rates or the yen on the back of Friday's BoJ decision. A hawkish Fed keeping USD/JPY bid in a broad 157-159 range into year-end is our current view.

One low-probability, high-impact risk for Friday is any announcement from Japan's GPIF pension on re-allocating funds domestically. We made the case in this article that a greater allocation to domestic securities would help the government's huge investment drive. One of the most memorable adjustments in those allocations was made on the same day as the BoJ's policy decision in October 2014.

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Fed Chair Kevin Warsh acknowledged that progress is being made, but inflation remains too high and a December FOMC rate hike remains our call

James Knightley, Padhraic Garvey, CFA and Chris Turner

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target rate range at 3.75%-4%. Only 2bp of a potential 25bp hike was priced ahead of the decision and only 6 of 97 organisations polled by Bloomberg predicted a hike, so this was no surprise and market reaction has been limited.

Senior Fed officials, including Vice Chairs Philip Jefferson and Michelle Bowman and Governor Chris Waller, suggested ahead of the decision that they were prepared to move cautiously, seeing little need for back-to-back rate hikes after September's policy change. Nonetheless, Beth Hammack and Lisa Cook both felt that the current situation warranted additional action. Cook's vote for a

hike was a surprise given the general sense that she was a centrist, and it will do nothing to improve her fraught relationship with President Trump. Instead, we had suspected it would be one or both of Neel Kashkari and Lorrie Logan (who both also dissented in July) that would join Hammack in voting in favour of a further rate hike.

There aren't many changes in the accompanying press release, with the economy described as "expanding at a strong pace" and "inflation remains elevated". There was no pre-commitment to further action, with it being the data that "will help the Committee determine the policy stance appropriate to ensure a timely return of inflation to the Committee's 2 percent goal."

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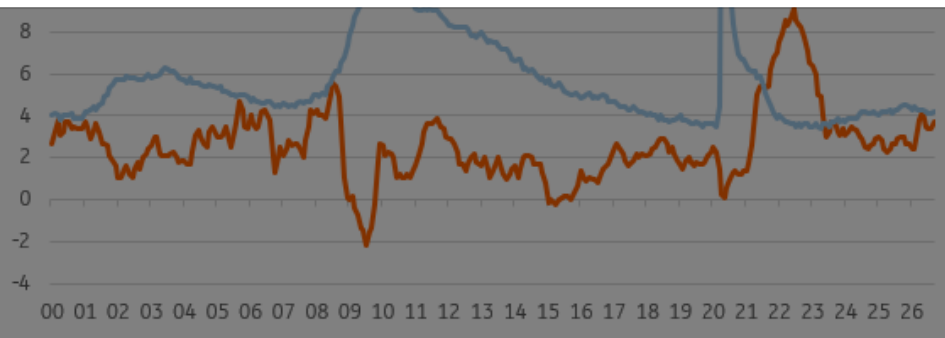

Source: Macrobond, ING

December rate hike remains the most likely path

There is the valid question that, given a majority of FOMC members project a rate hike before the end of the year, why didn't they take that decision today? Kevin Warsh's response was that he himself didn't make a prediction and, in any case, they removed some policy accommodation in September. Inflation is "going the right way" and he is encouraged by the progress made.

We will have two jobs reports and one inflation print ahead of the December decision. The November CPI will be released the day after the Fed meeting, and officials will almost certainly get to see it. As such, they will have more information than the market does, meaning a surprise can't be completely ruled out. For now, a D
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more disinflation, we believe overall US inflation can fall back to 2% in the second half of 2027.

Hence, we subsequently anticipate a prolonged pause through to December 2027, when we have that December 2026 hike being reversed. This is in line with the current consensus forecast, while the Fed's own projections (its central forecast) see one hike this year with rates left unchanged throughout 2027.

The lean hawkish is enough to validate the break back above 5% for the 10yr

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Hardly a word in the short statement that is not leaning hawkish. But no hike. Prior to the FOMC meeting, it was remarkable how the 10yr yield had been clinging to the 4.99% level. It was like a comfort level; technically below 5%, but not too far below in case it needs to turn tail and shoot back up again. It seemed that the US 10yr was waiting to tick off this outcome before deviating too much from 5%. Well, it's now had the outcome. The initial reaction was up. But the reaction to Chair Warsh's words at the press conference was to revert down to 5%, and briefly below. This can be rationalised by Warsh's comforting words on inflation expectations, which he deemed to have been tamed somewhat.

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likely still some way from the Fed spelling out a plan to shrink its balance sheet, which we know remains a policy objective for Chair Warsh. We expect recommendations from the various committees to surface closer to the turn of the year. Meanwhile, liquidity conditions are stable, with repo behaving far better than it did a year ago.

Dollar softens a touch

Perhaps it was a belated acknowledgement that there would not be forward guidance on a December rate hike, but the lack of clarity

there seemed to weigh on short-dated rates and the dollar. Chair Warsh actually sounded a little uncomfortable when a reporter asked whether the current policy setting was sufficient for the Fed to hit its goals.

The dollar reaction – a slight softening – probably owed to positioning. Here the consistent run higher in energy prices over recent months, solid US activity and French budget woes have largely seen one-way traffic in the dollar since early September.

However, the overall tone of the press conference, strong and broadening US growth and above-target inflation, suggests the d

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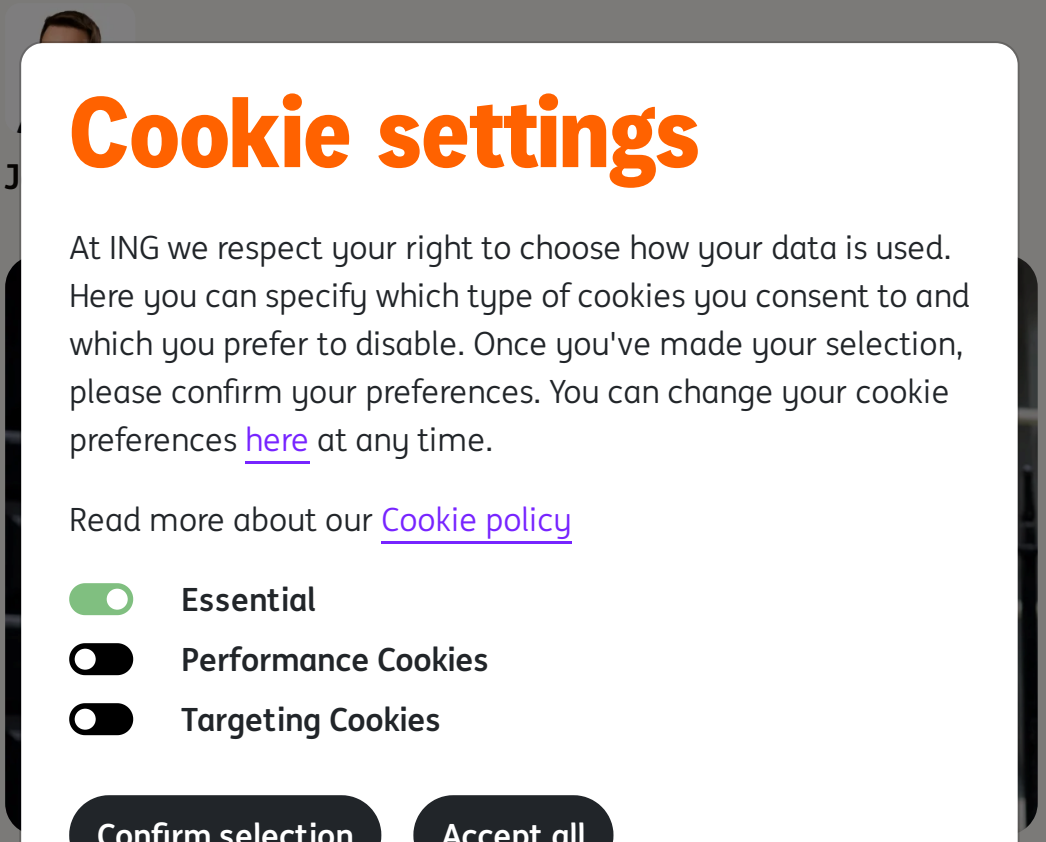
The UK's Autumn Budget keeps the deficit on a downward trajectory and, crucially, doesn't do anything that could make life harder for the Bank of England. But many of the UK's longer-term challenges remain unfunded and unaddressed, while the Budget keeps alive the option of an early election in 2027



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There's a lot to like for investors in today's budget, even if a lot of it had been trailed in advance. UK Chancellor John Healey has kept the long-planned fiscal consolidation on track, despite rising borrowing costs. And, crucially, he's avoided doing anything that could make life decisively harder for the Bank of England, with a rate hike likely just days away.

There are two key numbers. The first is the 2027 budget deficit, which the independent Office for Budget Responsibility forecasts at

3.4%. That's down from 4.2% this year, which, even after the Iran War, is itself down from 4.4% last year. That's after accounting for the rise in borrowing costs, which add around half a percentage point to the deficit each year.

The UK's fiscal consolidation remains in play. And though it's taking longer than previously expected, it still means Britain is the only G7 economy actually taking steps to improve its deficit.

The UK's deficit is set to fall, despite higher borrowing costs

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getting dragged into higher tax brackets. This 'fiscal drag' is driving the entirety of the deficit improvement next year; income tax as a share of GDP is projected to rise by almost half a percentage point. And aside from a small change to the basic allowance for low-income pensioners, there's nothing in Healey's first budget today that suggests this policy is imminently set to get watered down.

The result is that debt issuance – the number the markets ultimately care most about – is barely set to rise next year versus previous plans from the Debt Management Office. The £312bn gross financing requirement is below most people's expectations

(although there are a lot of moving parts here, and not all of them intrinsically linked to the government's fiscal choices).

The second key number is the degree of fiscal expansion. For the current fiscal year, this budget offers a small stimulus of roughly 0.2% of GDP – and for 2027, it's 0.1%. That's driven primarily by temporary and targeted energy support – a cut to VAT on electricity and a £100 rebate for certain households – which should also shave 0.1-0.2ppt off near-term inflation.

These are not the sorts of numbers that move the dial for the Bank of England. And though the BoE is on track to raise rates next week, a

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Source: Office for Budget Responsibility

Still, this is a budget that leaves more questions than answers.

How will the government account for higher inflation in departmental budgets? How will it fund defence? Both questions were left unanswered today. The government's overall spending pot for government departments was left largely unchanged later this decade, despite a higher path for inflation and the Chancellor's reaffirmed commitment to 3% NATO spending by 2030. Implicitly,

that pledge is being funded by squeezing other areas of government; the OBR's numbers imply real-terms cuts of 2.6% a year for unprotected departments later this decade.

That almost certainly won't happen – and this is before you take social care or any of Prime Minister Andy Burnham's other bold policies into account.

These thorny issues have been punted into 2027, potentially in the autumn, when the government will have to unveil its long-awaited Spending Review. This is the process that allocates spending across government departments and is fraught with political risk.

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European neighbours.

United Kingdom Budget

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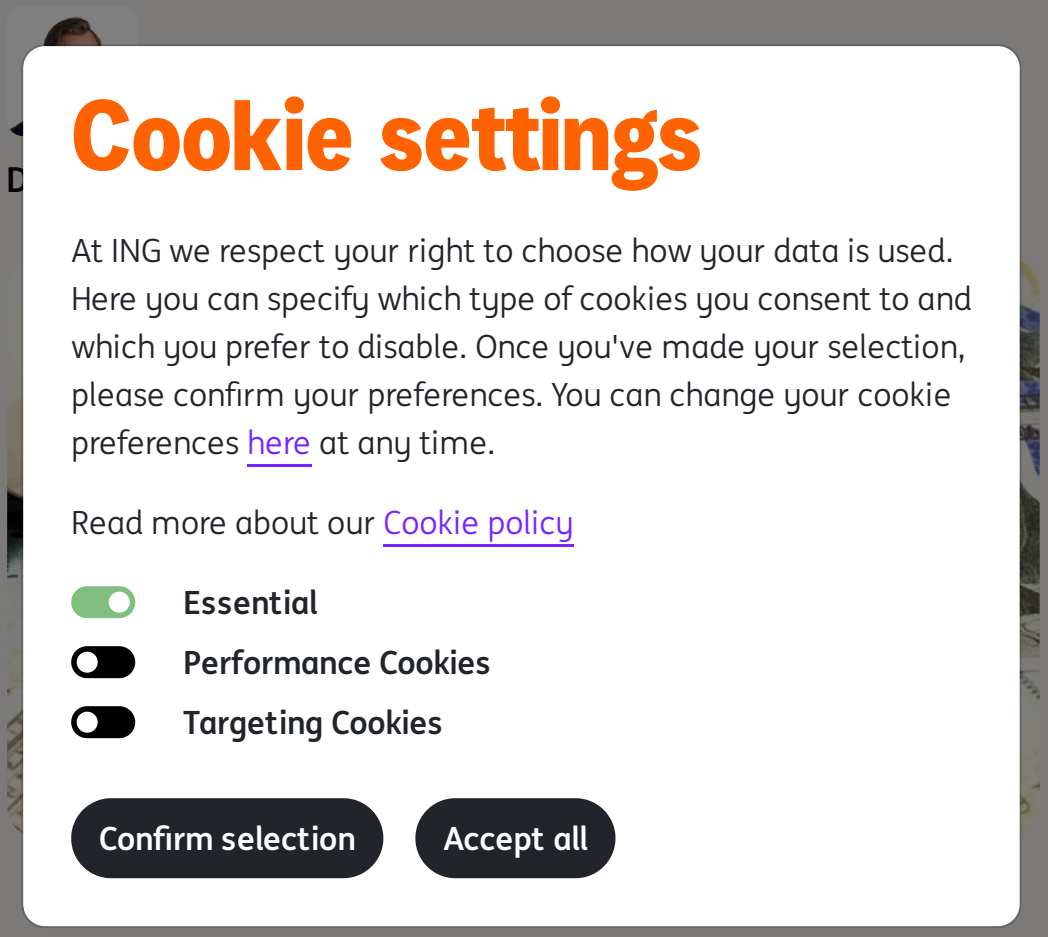
Rising fuel prices and the war with Iran have significantly increased the chances of a Democrat-controlled Congress. This could mean greater pricing pressure on pharma and more healthcare affordability measures. However, surprises do happen. We therefore assess three possible scenarios and what they mean for healthcare and pharma



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The US midterm elections take place on 3 November. All 435 seats in the House of Representatives and 35 of the 100 Senate seats are up for election. With Trump's approval rating hitting a record low according to a Fox News poll, it is no wonder that the pollsters currently suggest the Democrats lead by around 8 percentage points nationally. A Democratic takeover of Congress therefore seems most likely, but a Democratic clean sweep is not certain.

We look at what the possible outcomes could mean for healthcare and pharma.

The key differences

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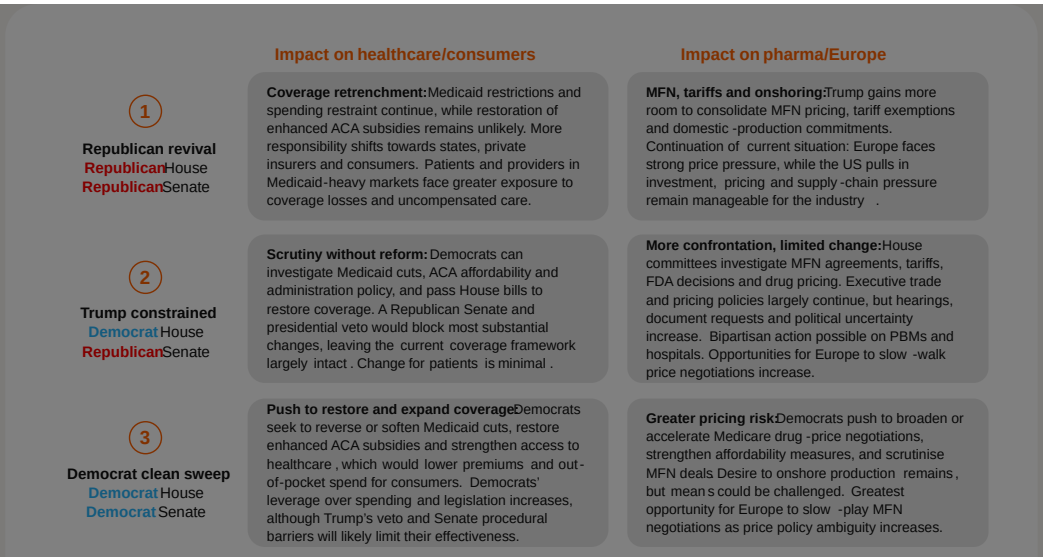
the objectives are largely bipartisan.

Midterm election scenarios and their impact on US consumers and Europe as well as healthcare and pharma



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greater scope to embed MFN pricing and use tariff exemptions and company agreements to further secure US manufacturing commitments. Pharmaceutical companies would gain greater clarity about the direction of policy, but not relief from pricing or supply-chain pressure. In short, this would be the scenario with the greatest continuity in pharma policy.

Scenario 2: Trump constrained

A Democratic House could obstruct legislation, control committees and launch investigations. But bills restoring ACA subsidies or

mitigating Medicaid cuts would face a Republican Senate and, ultimately, Trump's veto. Healthcare policy would therefore become more confrontational but not necessarily very different in practice. Coverage losses and stricter Medicaid eligibility rules would remain immediate issues, while Democrats would use hearings and must-pass spending legislation to put pressure on the administration.

Pharma would face detailed scrutiny of MFN deals, tariffs, and the implementation of Medicare negotiations. Yet neither party would control all the levers required for sweeping reform. This could create room for narrower bipartisan measures, particularly around Pharmacy Benefit Managers (PBM) and hospitals. Broader reform w… e
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w… negotiations agreed under the IRA, alongside greater scrutiny of launch prices and PBMs.

Moreover, stronger congressional protection of National Institutes of Health (NIH) funding, scientific institutions and Food and Drug Administration (FDA) review capacity would support biotech and early-stage innovation. If Democrats truly make drug-price negotiations a priority, then this scenario could lead to a mixed outcome: greater price pressure, but a stronger public research base and a better-funded FDA.

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Why a Democratic clean sweep would be good news for European governments

For Europe, pressure on the pharma industry to localise in the US will remain under all three scenarios. Democrats may criticise the means, but they also view pharmaceutical supply chains as a national-security issue. Republican and Democratic policy objectives therefore broadly align, while Trump retains substantial executive authority over trade.

The biggest difference is that a Democratic clean sweep would demand greater transparency around MFN deals and increase price p
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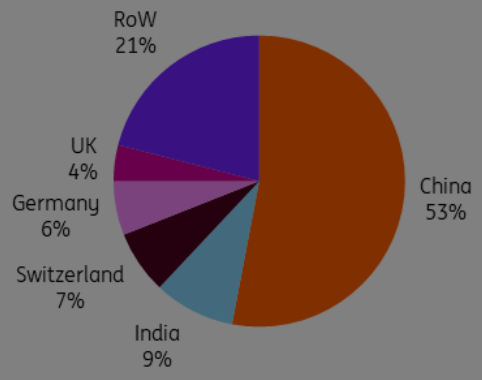
China will also keep facing bipartisan pressure. Both parties now view pharma and biotech as key to national security interests and support greater domestic manufacturing and reduced foreign dependencies. However, China's dominance in active pharmaceutical ingredients is unlikely to wane any time soon.

The US imports more than half of its pharmaceuticals from China

US pharmaceutical imports (branded and generic) by volume in 2023



Source: United Nations Commodity Trade Statistics Database; BMI; ING

Furthermore, commercial demand for Chinese innovation will not d
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US | Trump | Pharmaceuticals | Midterm elections | Healthcare | Elections

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Global air travel growth is set to rebound to 5% in 2027, but airlines face tougher conditions. Demand remains resilient, but the fuel price shock will spill over into 2027, while new aircraft supply remains constrained. This creates divergent fortunes across airlines, segments and regions

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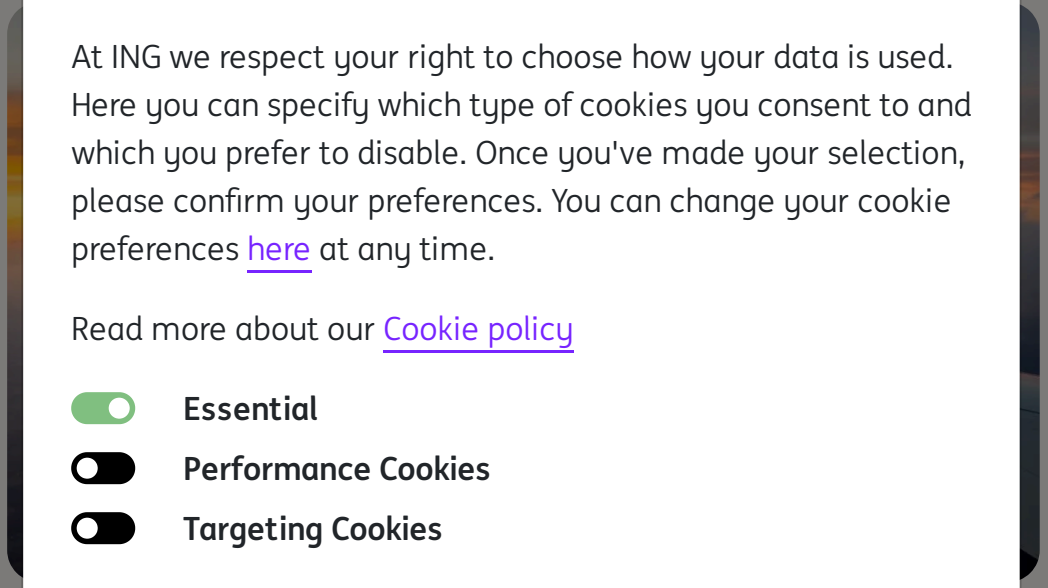

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- **The fuel-price shock will continue into 2027.** Jet fuel could account for close to 35% of airlines' operating costs in 2026, up from 27% in 2025. Tight refinery capacity and low inventories are likely to keep prices elevated even if oil prices ease.
- **European air fares will have further to rise.** Hedging initially protected many European airlines, but that cover will progressively expire during 2027. US fares have

already adjusted more quickly because American carriers tend to hedge less.

- **Budget airlines are particularly exposed.** Fuel represents a larger share of their costs, while price-sensitive passengers make it harder to pass on the full increase. Premium carriers have more scope to raise fares without significantly weakening demand.
- **Ticket prices are unlikely to return to early-2026 levels**. Even if fuel becomes cheaper, resilient demand, limited aircraft availability and wider inflationary pressures should keep fares relatively high.

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fleet could grow by approximately 8% a year over the next two decades.

Volume and demand developments in 2026 and 2027

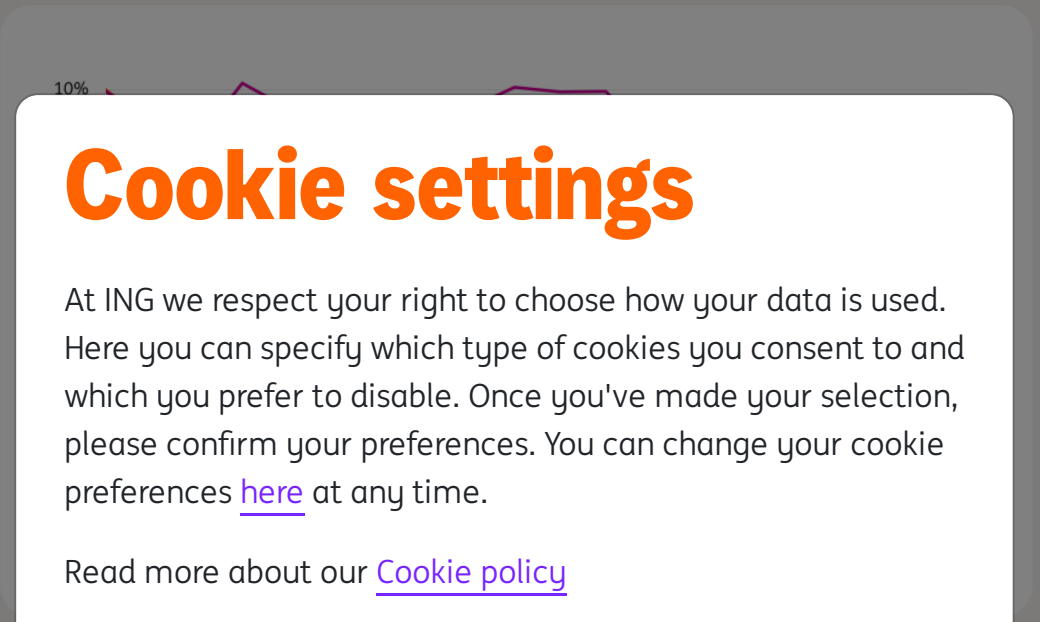
Airlines faced their most turbulent year since the pandemic and effects will linger

2026 has turned into the most turbulent year for the airline industry since the pandemic. Once again, pressure came from a

supply shock, with the Middle East war leading to disruption, including the blockade of the Gulf of Hormuz. This affected the industry throughout the rest of the year and its effects will spill over into 2027, too.

Middle East disruption wiped out growth but didn't derail global traffic

Monthly global passenger passenger km (PLF) in %



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And this comes on top of the Russian airspace, which complicates operations. Traffic in this pivotal region on the East-West corridor halved in March and April and was still 15% lower year-on-year in August 2026. Before the war, the Middle East accounted for 9.5% of global revenue passenger kilometres (RPKs). On balance, the setback in the region shaved some 2-3 percentage points off global growth with additional impacts elsewhere.

Despite a sharp increase in jet fuel prices, airline response remained constrained, with most measures limited to cancelling certain unprofitable routes rather than undertaking broader capacity

reductions. All in all, we still expect around 1% global passenger growth in 2026, but this obviously falls short of the structural 4-5%.

It remains unclear when sustainable peace and stability will return to the Middle East, but Gulf countries retain ambitious growth plans for hubs such as Dubai and Abu Dhabi, anticipating strong feeder growth from the expanding middle class in India, the world's most populous country.

Middle-East fall-out rattled the airline industry, but the impact on global passenger v

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but it needs to be profitable growth

Airlines enter 2027 in a much more difficult operating environment. Jet fuel prices remain elevated, geopolitical and security risks persist, and aircraft maintenance and delivery constraints continue to limit growth. Disruptions are also becoming more frequent, partly due to more extreme weather. In this environment, the performance gap between airlines is widening.

Yet air travel demand has proved remarkably resilient despite the turmoil and higher ticket prices. You might think geopolitical and safety concerns would keep travellers on the ground, but it appears

they more often opt to switch destinations. In Europe, flight traffic had already moved out of contraction before the summer, pushed by countries like Spain and Italy. This resilience was evident across all global regions outside the Middle East, with particularly strong growth in emerging aviation markets such as Africa and Latin America.

Economic growth is expected to pick up in 2027, while the propensity to travel remains strong. Against this backdrop, we expect global passenger growth to rebound to around 5% in our base case.

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Source: IATA, *ING Research (base case)

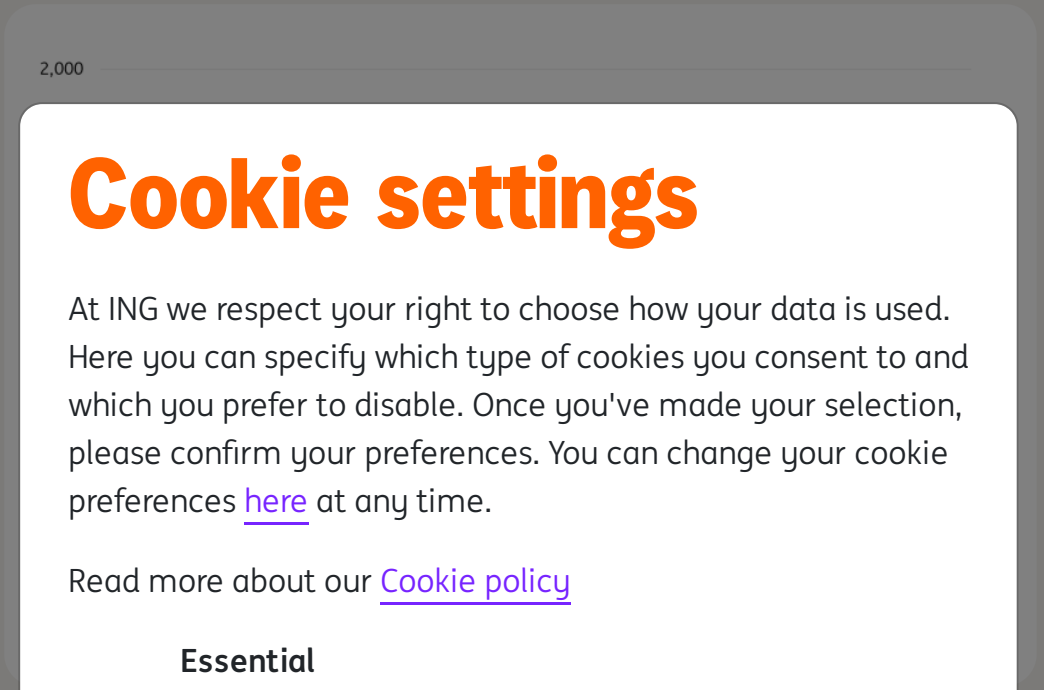
European carriers benefited from the setback on the Middle East-Asia corridor

Travel patterns changed after the Iran war started, while Middle Eastern carriers and transfer hubs were hit hardest by airspace closures. European carriers such as Lufthansa and Air France-KLM benefited from the disruption, partly cushioning the impact of surging fuel prices.

Traffic in the Europe-Asia corridor was still up more than 12% year-on-year in August, while the Middle East-Asia corridor was down almost 12% following much steeper declines in March-May. But with the conflict persisting and renewed security incidents, the impact could extend into 2027.

Jet fuel continues to trade high in Q4 2026

European jet fuel prices in $ per tonne (ARA)



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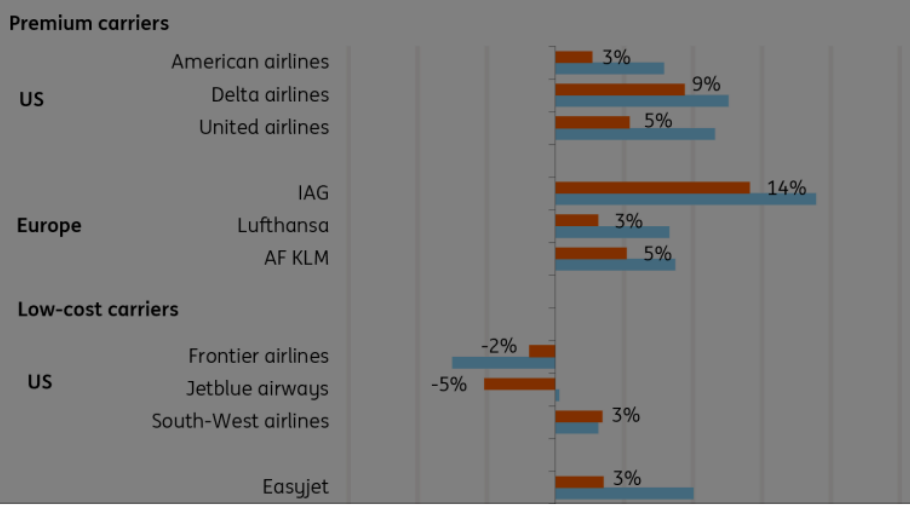
This share could move towards 35% in 2026 as a result of persistently high jet fuel prices and elevated crack spreads, caused by the loss of Middle Eastern supplies and tight global availability.

Further geopolitical developments remain highly uncertain. Even if the Middle East conflict is resolved and oil prices retreat, tight refinery capacity and depleted inventories are likely to keep middle-distillate prices, including jet fuel, elevated well into 2027.

High fuel prices impact profits in US and Europe, low cost carriers most vulnerable

Operational profit (EBIT) in % per peiod



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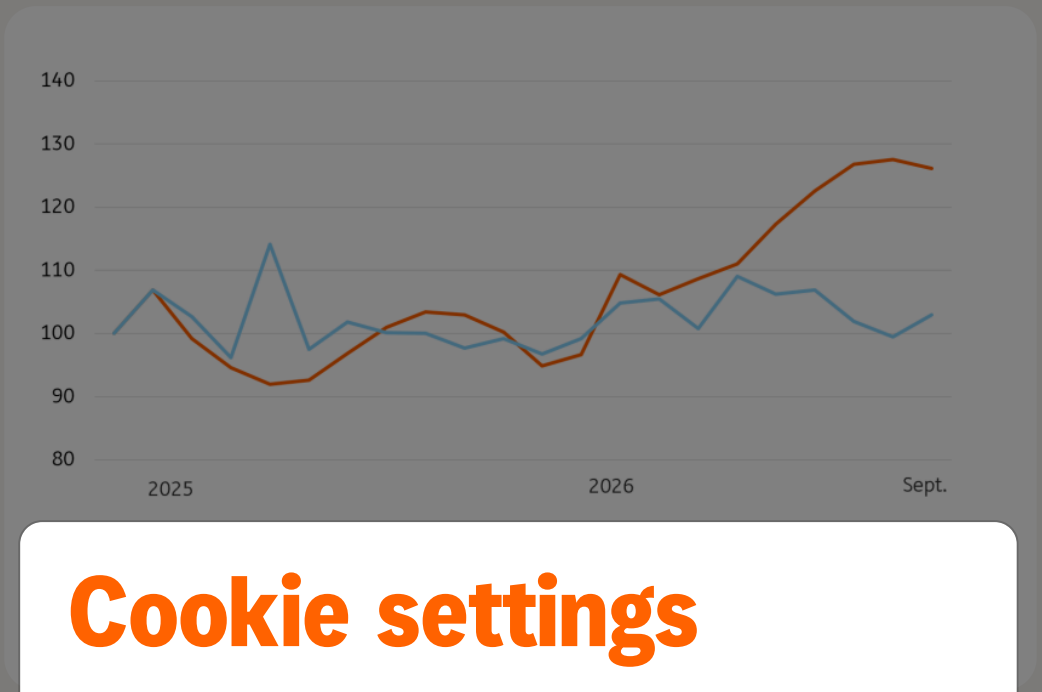
exposure to prevailing market prices. Ryanair, which remains (80%) hedged until the second quarter of 2027, has already warned of significantly higher fares.

Outside, the picture is mixed, but hedge ratios are generally lower, and airlines in the US and China, for instance, don't hedge at all. This means that two airlines operating similar routes can face very different fuel bills.

Airline tickets in the US responded much faster to the cost shock than in Europe

Index ticket prices vs consumer price index (CPI, seasonally adjusted) in the US and EU (Jan 2 = 100)



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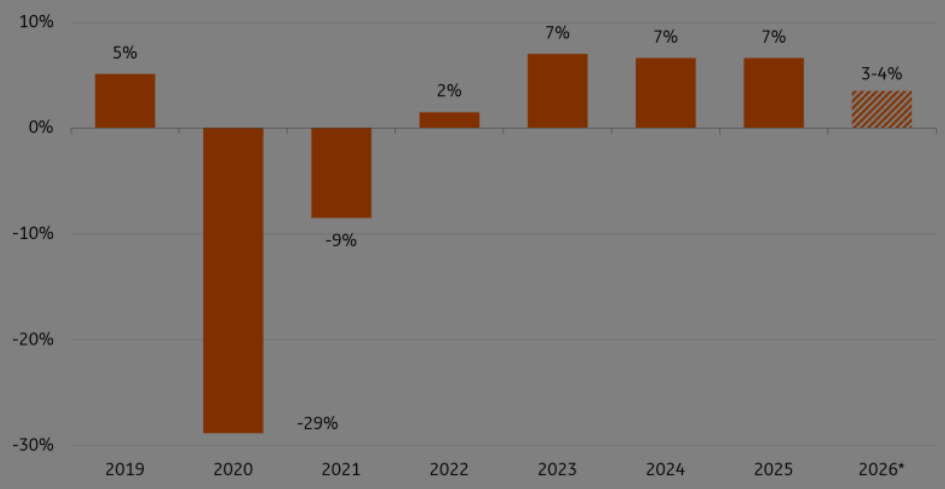
significantly if fuel prices fail to fall. Coverage gradually expires as 2027 progresses, leaving airlines increasingly exposed to higher spot prices.

Airline profits hit, but industry remains profitable

Global development of operational profits (EBIT) per year



Source: IATA, *ING Research based on IATA

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adds some $300 per tonne of jet fuel in 2026 (BNEF, ING Research).

Meanwhile, France and the Netherlands have either raised or plan to raise ticket taxes, particularly on long-haul flying, while Germany has scaled its levy back and Sweden has abolished it.

*A revision proposal is in progress for aviation, drawing a circle of 5,000 km around Frankfurt for application of ETS, which would hit Middle East airlines most as they will then be in scope. **In 2025, the blend rate stood at 2.8%

High fuel prices put more pressure on low-cost carriers

Low-cost carriers are particularly exposed because fuel accounts for a larger share of their cost base, and profit bases are more vulnerable. High hedge ratios initially cushion volatility, but these hedges gradually roll off. The more challenging environment is already visible in the performance of several European low-cost carriers and will remain an important theme in 2027 if fuel prices stay elevated.

Premium carriers have more room to manoeuvre. Demand for premium seats (business, premium economy) is less price-sensitive, providing airlines with greater scope to pass on higher costs. Both United Airlines and Delta Airlines reported little impact a

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profits in the global industry are expected to drop meaningfully below 5%. Airlines including Ryanair, Southwest Airlines and AirAsia have scaled back capacity on unprofitable routes as elevated fuel prices persisted longer than expected.

According to global aviation consultancy Cirium, around 1% of global air travel capacity will be removed heading into the Northern Hemisphere winter of 2026-27. In less mature markets, maintaining volumes while passing on the higher fuel bill is more difficult. Airlines may also prioritise less fuel-efficient, older-generation aircraft over capacity growth.

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Industry headwinds could become a catalyst for further consolidation, ranging from minority stakes to full takeovers.

Fares could retreat if fuel prices ease, but don't expect too much

Dynamic and algorithm-based pricing is becoming increasingly common. This helps airlines optimise already historically high load factors (84% in 2026 so far) but can also contribute to greater fare volatility, especially as passengers book later amid uncertainty.

If fuel prices ease in 2027, fares could drop back. Yet relatively

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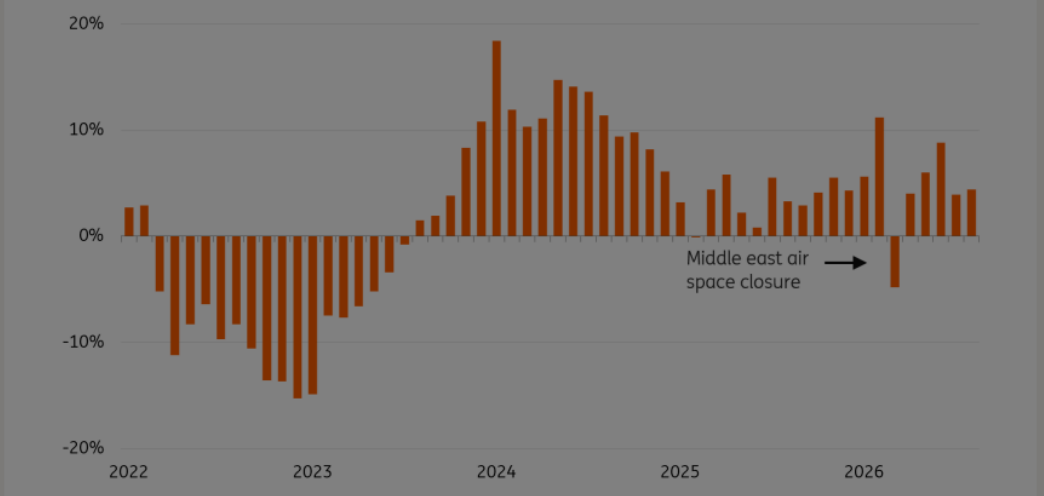
Freighter capacity received a boost during the pandemic through passenger-to-freighter conversions, but that boost has largely run its course. New-generation A350 and Boeing 777 freighters should add capacity in the coming years.

Aircargo volume resilient in disruptive geopolitical environment

Global development of air cargo volume in tonne/km (CTK) y.o.y.



Source: IATA, ING Research

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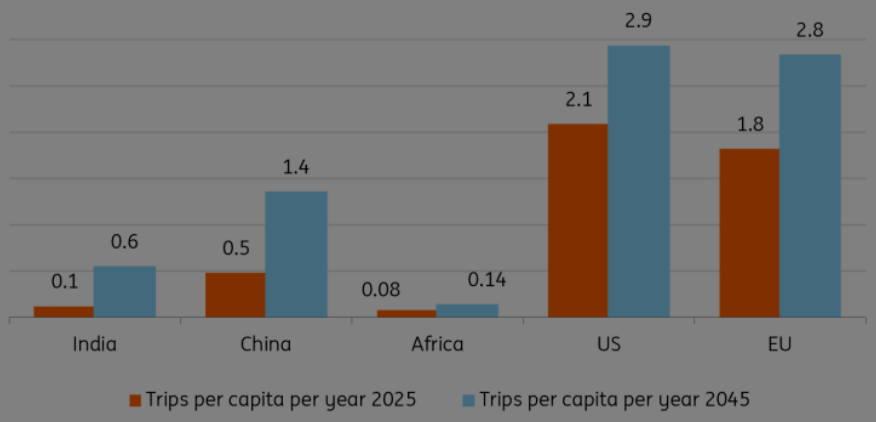
People still expected to fly more, especially in India

Trips per capita per year in various countries



Source: S&P, Airbus, ING Research

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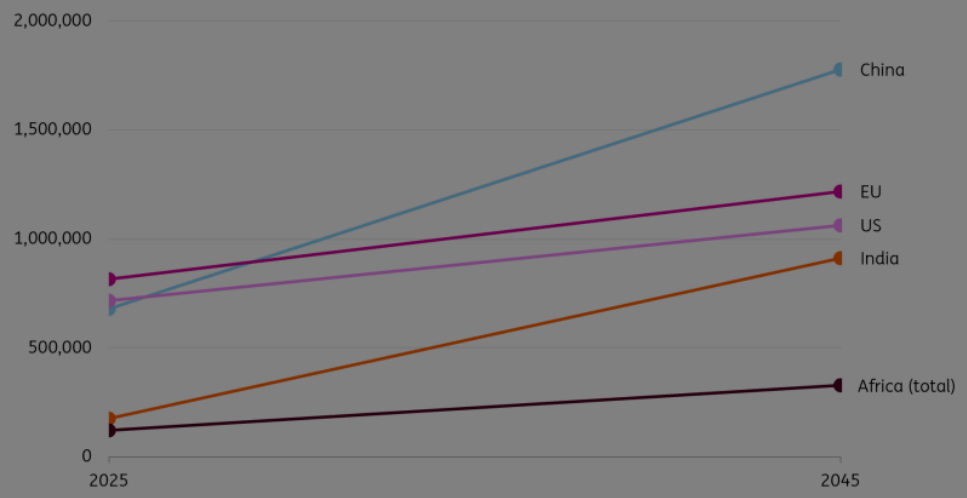
expected to grow by around 8% annually on average over the next two decades.

China and India are the growth engines of global aviation in the decades ahead

Forecasted growth in number of trips per year, per country/bloc (*1,000)



Source: S&P Global, Airbus, ING Research

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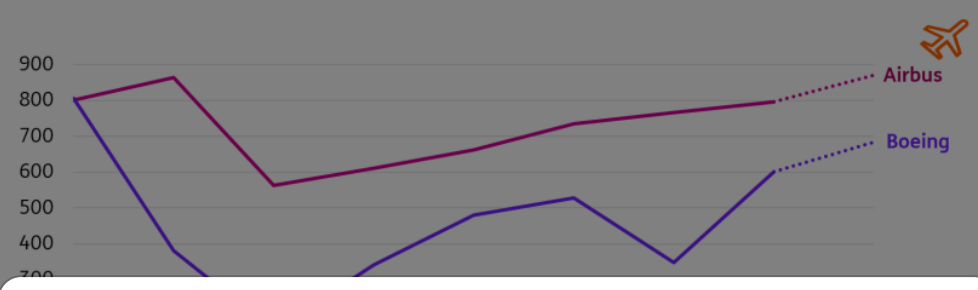
15,000 aircraft by mid-2026. Production and deliveries continue to lag order intake after seven consecutive years of subdued deliveries from 2019 onwards. Demand for new-generation aircraft remains strong, as illustrated by major recent orders from Avolon and Turkish Airlines.

Production could finally return to pre-pandemic levels in 2027. Chinese manufacturer COMAC, meanwhile, is making progress in China in Asia, but Western certification remains distant.

New airplane deliveries continue on a recovery track now with Boeing rebounding

Number of delivered commercial airplanes by manufacturers Airbus & Boeing per year



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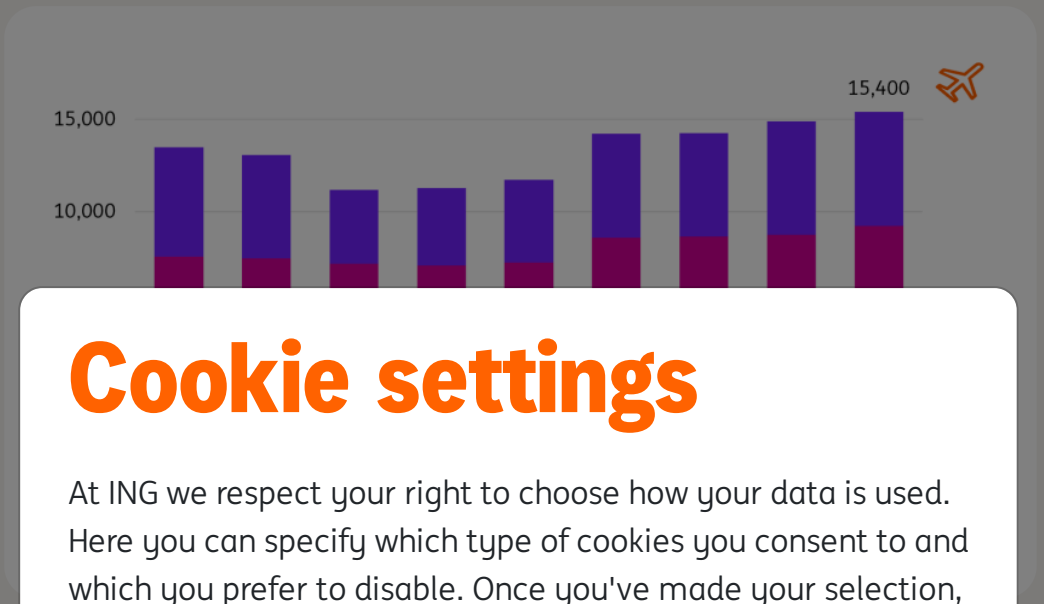
requirements mean manufacturers rely on a limited number of specialised suppliers. Engines and sophisticated cabin components still have long lead times, while skilled workers remain scarce.

- **Engine issues continue to affect available capacity.** The peak impact of the Pratt & Whitney inspection programme after an identified production issue has passed, but checks and maintenance continue to constrain parts of the A320neo-family fleet and absorb scarce maintenance capacity.

Aircraft production still has a long way to go catching up: backlog hit fresh record in 2026

Orderbook Airbus & Boeing: number of commercial airplanes



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scarce; they also consume significantly more fuel and therefore emit more.

Boeing is finally regaining altitude supporting deliveries

Boeing is finally making a sustained comeback following an unprecedented period of disruption stemming from the 737 MAX accidents, the pandemic and subsequent production and quality problems. It received certification for its 737 MAX 7 and also the 737 MAX 10 recently. Boeing plans to increase production for the family from 38 to 47 a month.

Higher 737 production should support deliveries in 2027 and gradually bring pre-pandemic production levels back into sight. This is hugely important for Boeing-dependent airlines such as Ryanair and Southwest. Progress on certification of the 777-9 is also important for long-haul fleet renewal.

Fleet renewal is currently the most important decarbonisation lever

Air travel continues to grow, and associated emissions tend to continue to rise further as well. Sustainable aviation fuels are labelled as an important industry pathway to decarbonise, but g

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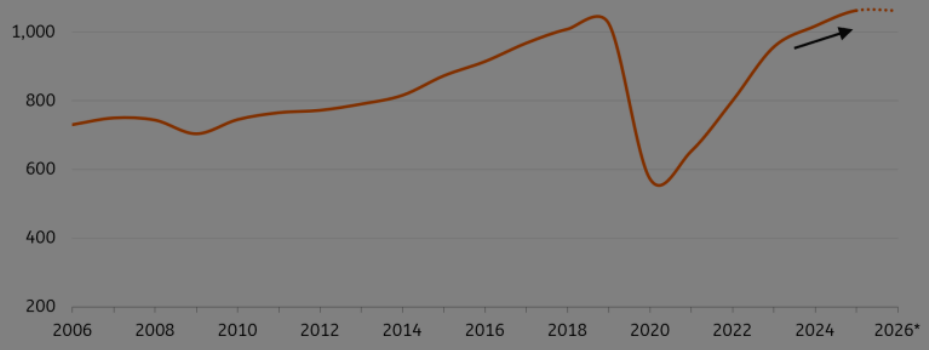

Source: IEA, ING Research *estimate forecast

The challenge is to deliver profitable growth in a more disruptive world

For airlines, the challenge heading into 2027 is therefore not simply finding sufficient demand. Relatively poor punctuality, with only around 61% of flights departing on time in 2026 year-to-date, is a signal.

The globally operating industry faces more turbulence and uncertainty amid intensified geopolitical risks. The challenge is to deliver profitable growth in a world of persistently constrained capacity, higher costs and more frequent disruption.

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CPB Netherlands Bureau for
Economic Policy Analysis

National Productivity Board 2024 annual report

Labour productivity in the market sector declined by 1.4% in 2023.

Exporters are the most productive firms in the Netherlands, but also their suppliers perform above average.

CPB - February 2025

Leon Bettendorf &
Michael Polder (Statistics Netherlands)

Preface

Productivity has shifted to the top of the political agenda in the last years. In view of the limitations to a further expansion of the labour supply, productivity growth is the principal way to support our future living standard. Unfortunately, the Netherlands, as all OECD countries, has been struggling with a declining trend in productivity growth during the last decades. We do not yet fully understand the causes of this global productivity slowdown.

CPB is part of a network of independent National Productivity Boards in the EU. The CPB has been a National Productivity Board since 2017. The main NPB tasks are monitoring and analysing productivity developments in the Netherlands. As in previous years, we summarise research on productivity issues in this Annual Report 2024.

We are preparing three publications this year. First, we participate in a multi-country project of the Productivity Institute (UK). The aim of this project is to identify and compare country-specific productivity trends and pro-productivity policies. Second, we will analyse the evidence on changing business dynamics in the Netherlands and the consequences for firm-level productivity and innovation. Finally, we will provide the analysis underlying our projections of long run productivity.

Pieter Hasekamp
Director CPB

Samenvatting

De arbeidsproductiviteit in de marktsector daalde in 2023 met 1,4%. Sinds 2000 is de productiviteit alleen in 2009 nog sterker gedaald. De uitfasering van de gaswinning sinds 2014 heeft de jaarlijkse productiviteits-groei met gemiddeld 0,3%-punt gedrukt.

We begrijpen nog niet helemaal waarom de productiviteitsgroei is afgenomen. Zoals in alle OESO-landen heeft Nederland last van een vertraging van de productiviteitsgroei. In een literatuuronderzoek vinden we verschillende verklaringen, maar de bijdrage van elke oorzaak is moeilijk te bepalen vanwege de sterke verbanden tussen de oorzaken.

Het productiviteitsverschil tussen de 10% meest productieve bedrijven en minder productieve bedrijven nam in 2021 toe. We stelden in ons vorige rapport (CPB, 2024) vast dat de productiviteitskloof tussen de bedrijven van de top 10% en de bedrijven van de onderste 10% sterk is toegenomen in 2020. Uit een update van deze analyse blijkt dat de kloof in 2021 groot bleef en niet te wijten was aan de bijzondere omstandigheden in 2020. Productiviteitsgroei in de economie kan voortkomen uit productiviteitsgroei binnen bestaande bedrijven of door activiteiten te verschuiven naar productievere bedrijven. De bijdrage van de verschuiving daalde in de verwerkende industrie en de dienstensector in de periode 2011-2019. De bijdrage van de bedrijvendynamiek bleef stabiel in de handel, terwijl deze belangrijker werd voor overige industrieën (bouw, energie, water en afval).

Exporteurs en hun directe toeleveranciers vormen de meest productieve groep bedrijven in Nederland. Uit de literatuur weten we dat exporterende bedrijven productiever zijn dan niet-exporterende bedrijven. We laten met nieuwe netwerkdata een duidelijk patroon zien: exporteurs zijn groter en productiever dan hun toeleveranciers, die op hun beurt beter presteren dan hun eigen toeleveranciers. De productiviteit van exporteurs en hun toeleveranciers wordt versterkt via drie mechanismen: leereffecten, selectie en schaalvoordelen. Beleid dat bedrijfsnetwerken versterkt, kan bijdragen aan productiviteitsgroei.

Summary

Labour productivity in the market sector declined by 1.4% in 2023. Since 2000, only 2009 saw a stronger fall in productivity.. Phasing out gas extraction since 2014 depressed annual productivity growth by 0.3%-point on average.

We do not fully understand why the productivity growth has decreased. As in all OECD countries, the Netherlands is suffering from a slowdown in productivity growth. A survey of the literature provides various explanations, but the contribution of each cause is difficult to determine due to the strong links between the causes.

The productivity difference between frontier firms and non-frontier firms was growing in 2021. We observed that the productivity gap between the top-10% firms and the bottom-10% firms strongly increased in 2020. An update of this analysis shows that the gap remained large in 2021. Productivity growth can originate from growth within existing firms or by reallocating activities to more productive firms. The contribution of reallocation declined in manufacturing and services during the period 2011-2019. The contribution of business dynamics remained stable in trade while it became more important for other industries (construction, energy, wate and waste).

Exporters and their direct suppliers form the most productive group of firms in the Netherlands. From previous research we know that exporting firms are more productive than non-exporting firms. From new network data, a clear pattern emerges: exporters are larger and more productive than their suppliers, who, in turn, outperform their own suppliers. Productivity of exporters and their suppliers benefit through three main channels: learning effects, selection and economies of scale. Policies that strengthen firm networks can contribute to fostering productivity growth.

1 Introduction

This Annual Report summarises research on productivity developments in the Netherlands.[1] It consists of four parts.

In the next section we summarize the unfavourable productivity developments in 2023. We show that phasing out gas extraction since 2014 contributed to the low productivity growth rates in the Netherlands compared to other European countries.

Various explanations are found in the literature why productivity growth is slowing down. No consensus is yet reached on the contribution of each explanation of the slowdown. We discuss the main studies in the third section.

The fourth section, provided by Michael Polder (Statistics Netherlands), updates and extends the evidence on firm-level productivity. He documents productivity differences between firms and shows that the contribution of continuing firms, new firms and closing firms to productivity growth differ across industries, as well as between time periods.

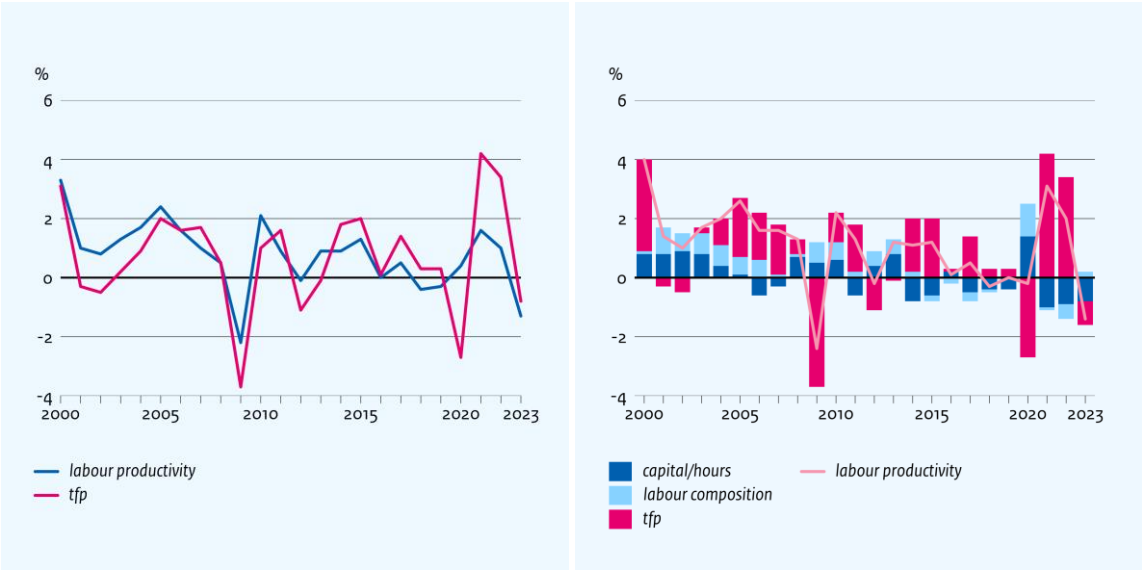
The last section exploits new data on the Dutch production networks.[2] It is well known that exporting firms are more productive than an average firm. This section discusses the impact on productivity of firms that are suppliers to exporting firms.

[1] Our Annual Report of 2023 can be found at annual report 2023. The reports of the other NPBs are collected at this EC-website.
[2] This is a translation of Freeman et al. (2024).

2 Recent productivity developments

Labour productivity in the market sector in 2023 was 1.4% lower than in 2022. Labour productivity is defined as value added per hour worked.[3] Figure 2.1a shows that productivity only declined more (with -2.2%) during the great recession in 2009 (CBS, 2024). The decomposition in figure 2.1b shows that the negative labour productivity growth in 2023 was driven by negative contributions of TFP growth (-0.8%) and capital deepening (-0.8%).[4] The number of working hours has increased, while the capital stock reduced (due to non-ICT capital). The long term decline in capital stock per hour since the mid 2010-s seems to continue at an even faster pace. TFP growth (a measure of the efficiency with which labour and capital are used in the economy) turned negative in 2023. TFP grew fast in 2021 and 2022, in part through recovery from the corona pandemic shock in 2020

Figure 2.1 Labour productivity in de market sector declined in 2023



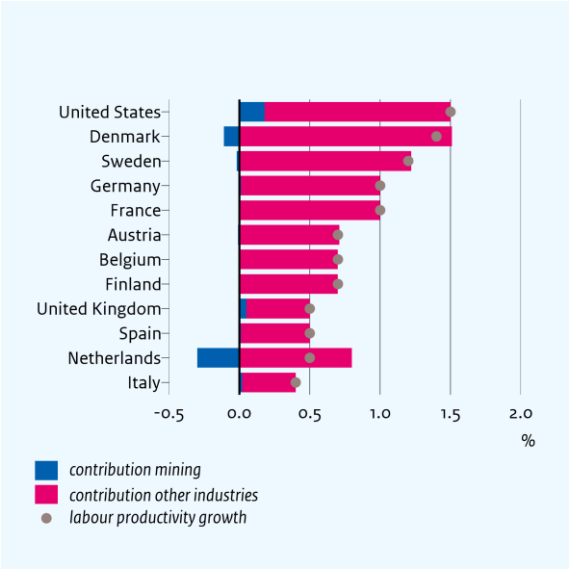
Source: CBS (2024); 2022 & 2023 are preliminary.

Phasing out gas extraction depressed productivity growth, starting in 2014. The mining and quarrying industry is very capital intensive, resulting in a high labour productivity. Labour productivity in this industry grew on average with 0.7% before 2014, while it fell with almost 16% per year after 2014 due to the phasing out of gas extraction (De Vries et al., 2024). Figure 2.2 shows that the average labour productivity growth in the Netherlands was one of the lowest in Europe. However, whereas the average contribution of the mining industry was negligible in most of the countries, it was negative (-0.3%-point) in the Netherlands (and +0.2% in the US). The productivity growth of the market sector excluding mining in the Netherlands was much closer to the average value in Europe. Next to mining, other industries that contributed negatively to productivity growth in 2023 are transport and storage (-0.5%-point) and manufacturing (-0.4%-point).

[3] The market sector includes all sectors, with the exception of the government, education, households and rental & trading of real estate.

[4] We use the identity: $\tilde{Y}_t - \tilde{H}_t = \alpha_t(\tilde{K}_t - \tilde{H}_t) + (1-\alpha_t)\tilde{C}_t + \tilde{A}_t$, where $\tilde{Y}$ is value added growth, $\tilde{H}$ is growth in hours worked, $\tilde{K}$ is capital growth, $\tilde{C}$ is labour composition growth, $\tilde{A}$ is TFP growth and α is the share of capital costs.

Figure 2.2 Average annual growth of labour productivity in the Netherlands was low due to mining (2013-2019)



Source: De Vries et al. (2024), Figure 4.2 & 5.1

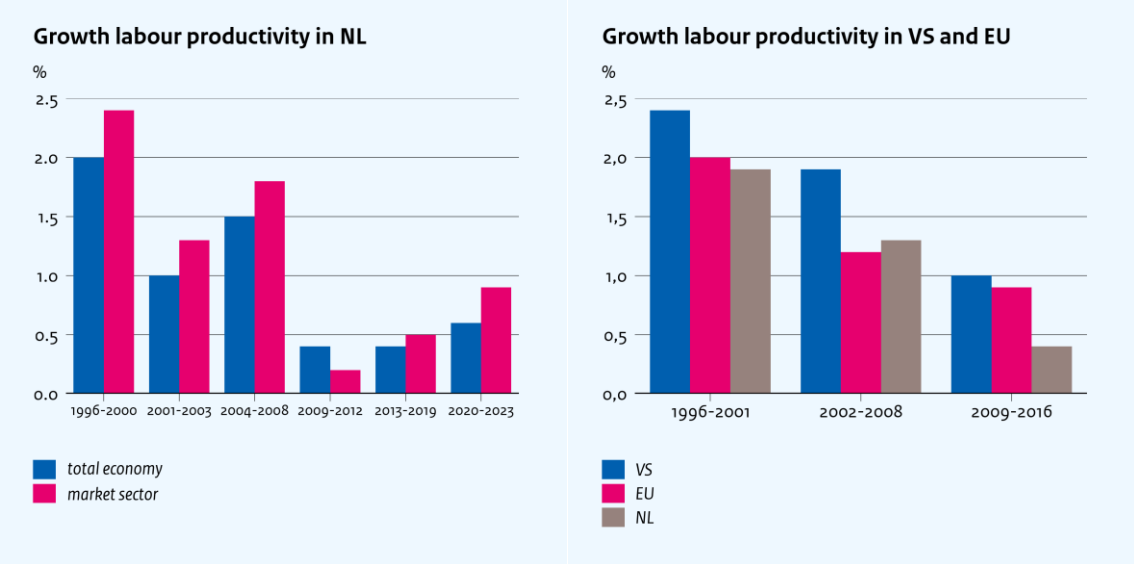
3 Explaining the productivity slowdown: a survey

We do not sufficiently understand current productivity developments, which makes insight into future developments uncertain. As in many other countries, the Netherlands is suffering from a slowdown in productivity growth. In the literature, various explanations are put forward as to why growth has decreased, but the contribution of each cause is difficult to determine due to the strong links between the causes. In addition, it is uncertain whether the outlined trend developments are permanent or temporary. For example, will artificial intelligence really develop into a general purpose technology that will put an end to the weak productivity growth? And, if so, how long will that last?

All OECD countries are experiencing slowdowns in productivity growth. Figure 3.1a shows the annual growth of labour productivity in the Netherlands in the total economy and in the market sector (this is excluding the government). GDP per hour worked grew on average around 2 percent around the turn of the century. This growth has fallen back to below 1 percent after 2009. Figure 3.1b shows the slowdown in other countries. The decline in GDP growth per hour worked started in all countries before the Great Recession (2008-2012). The Great Recession may therefore have exacerbated the slowdown but cannot be the cause (Fernald et al., 2023).

The global slowdown in productivity growth suggests common causes but there is no consensus yet on which cause is the most important and not even on whether the slowdown will be temporary or long-term. This slowdown puzzle contributes to the great uncertainty of projections in the near and distant future. We discuss the most important, global causes one by one, although they are strongly related. [5]

Figure 3.1 Labour productivity growth is falling (value added per hour worked)



Source: De Vries et al. (2024), figure 3.1.; De Bondt et al. (2021), figure 3.1.1.

[5] See the surveys of Moss et al. (2020), Akcigit et al. (2021) and Goldin et al. (2024). For a discussion of specific Dutch causes, such as the large share of self-employed people and low R&D investments, we refer to Roelandt et al. (2019) and de Bondt et al. (2021).

3.1 The diminishing potential of innovative ideas

Technology pessimists are skeptical about innovations that can still significantly increase productivity. Gordon (2016) argues that the number and breadth of innovations was exceptional in the period 1870-1970 and that we have now returned to the normal, structural level. Bloom et al. (2020) argue that the productivity of researchers has fallen sharply because it is becoming increasingly difficult to find new ideas. They express the TFP growth as the product of the number of researchers and their productivity. They observe a strong increase in the effective number of researchers, but the problem is that productivity in research has fallen just as sharply. The best illustration is provided by Moore's Law. The number of transistors in an integrated circuit has doubled every two years, but this required an increasing input of resources.

Jones (2022) lists three reasons why growth in GDP per capita will slow down in the future. Jones analyses economic growth in the US with a model in which TFP growth is determined by a Research & Development (R&D) sector. The first reason for a slowdown in growth is that the three main determinants after 1953, apart from population growth, can hardly grow any further. The contribution of population growth to growth in the past is 20%, while 80% stems from a rising level of education, an improved allocation of production factors and a growing size of the R&D sector (as % of population). The problem is that the last three factors are naturally limited. The second reason is that a slowdown in employment growth in R&D is already observed. The third reason is that natural population growth (excl. migration) is already negative in many developed countries. In the long run, economic growth remains proportional to population growth.

Technology optimists, on the other hand, emphasise the future potential of innovation in ICT; see for example Brynjolfsson and McAfee (2014). As with previous industrial revolutions, they expect that it will take time before the combination and diffusion of multiple new technologies will lead to a revival of productivity growth. Artificial Intelligence (AI) could become a new general purpose technology. AI is expected to have major effects on the economy, and on productivity in particular, because AI not only increases the productivity of the production of products but also the production of innovative ideas (R&D). Initial studies estimate that AI increases macroeconomic productivity by 1%-point per year for 10 years (see the discussion in Aldasoro et al., 2024). However, Acemoglu (2024) is skeptical about the extrapolation of early, simple applications and calculates that TFP growth increases by less than 0.1%-points per year. In addition to the potential of AI, Jones (2022) believes that growth can accelerate in the future because there is still sufficient room for expansion of the R&D sector at the global level.

Philippon (2022) disputes the existence of a structural break in technological progress. He criticizes the standard specification of the TFP growth rate. The literature assumes a constant TFP growth rate (i.e. exponential growth), while Philippon argues that the evidence is more in line with a constant absolute growth rate (additive growth). If absolute growth remains constant, the TFP growth rate will decline in the future. In the US, he finds a structural break in TFP growth around 1930 in the period 1890-2019. Due to catch-up growth, TFP growth in the eurozone and Japan was higher than in the US until 1990. After 1990, absolute growth declined sharply and became smaller than in the US. Assuming additive growth, Philippon finds no evidence of a recent slowdown in technological progress.

3.2 Divergence between high- and low-productivity firms

The OECD found that the difference in productivity between high-productivity firms and low-productivity firms has increased (Berlingieri et al. 2017a). Because knowledge about new production techniques and organizational forms is spread less quickly, low-productivity firms are less able to keep up with the

productivity growth of high-productivity firms. The deterioration in knowledge diffusion might be related to the increasing importance of intangible capital (see section 3.6). The divergent productivity development may explain the slowdown in productivity growth.

In the Netherlands, the productivity difference between the most productive and median firms started to increase around 2015. De Bondt et al. (2021, Table 5.3.2) find that the difference in both labour productivity and TFP increased before 2005 during the period 2000-2015. The divergence in productivity was stronger in industry than in the service sector. Research in van Heuvelen et al. (2018) confirms that the productivity gap between the top 10% firms and the median firms has remained stable in the period 2006-2015. In the next section we show with a new dataset that the top-median gap increased by 0.7% annually from 2015 to 2019. The divergence hypothesis is not a good explanation since the slowdown in productivity growth started much earlier in the Netherlands.

3.3 Increase in market concentration

According to this hypothesis, lower productivity growth is associated with the trend towards less competitive markets. Firms with more market power have fewer incentives to increase their productivity. The emergence of less competitive markets is related to the superstars hypothesis (or winner-takes-all).[6] After superstars have acquired a dominant market position, they shift their activities from innovative investments to strategies to protect their excess profits. De Ridder (2024) argues that the increase in intangible capital explains these developments. Since intangible capital is characterized by a combination of low marginal costs and high fixed costs, this promotes the growth of large firms and a reduction in productivity growth. The increase in market power seems to be a problem mainly in the US.[7] De Loecker et al. (2020) estimate the markup of firms (defined as the ratio of price and marginal costs) as a measure of market power. They find that the average markup in the US nearly tripled from 1980 to 2016.

However, an increase in market concentration can also promote productivity growth. When more productive firms gain market share at the expense of less productive firms, the reallocation contributes to aggregate productivity growth.[8] In Europe, the increase in market concentration does not seem to be associated with more abuse of market power. Bighelli et al. (2023) estimate stable markups in 15 European countries. Van Heuvelen et al. (2021) do not find that markups in the Netherlands increased significantly in the period 2006-2016. The slowdown in productivity growth in the Netherlands therefore does not seem to be explained by the increase in the market power of firms.

3.4 Weakening of business dynamism

Productivity growth has been affected by reduced business dynamism. With healthy business dynamism, old, low-productivity firms are replaced by more productive new firms (principle of creative destruction). Released inputs are put to more productive use, which contributes to a stimulation of productivity growth. Decker et al. (2017) point to the relationship between reduced business dynamism and slowed productivity growth in the US. Freeman et al. (2021) show that business dynamism in the Netherlands is slowing down (after 2006), mainly as a result of a declining share of start-ups. They find that the reduced dynamism has

[6] See e.g. Autor et al. (2020).
[7] See e.g. Eeckhout (2022). This hypothesis is contested by Shapiro et al. (2024).
[8] See the discussion of the distinction between good and bad concentration in Covarrubias et al. (2019).

contributed to lower TFP growth (by 0.2%-points per year), especially in the service sector. The weakening of business dynamism may be a cause, but not the main cause of the slowdown in productivity growth.

3.5 Shifts in sector structure

The shift towards the service sector reduces average productivity growth. The share of the service sector (incl. healthcare) in the economy increases, while the productivity of the service sector grows less than that of industry. De Bondt et al. (2021, section 5.2) confirm that the shift towards the healthcare and service sectors has had a negative effect on productivity growth in the period 1996-2016, but that this shift is not large enough to really explain the slowdown in productivity growth (see an update in De Vries et al., 2024). Most of the productivity growth is found within the industries, while the negative reallocation effect between sectors remains small. Duernecker et al. (2023) compare the strong growth in labour productivity in the period 1970-1985 with the slow growth in 2002-2017 for several high-income countries. They also show that the decline in productivity growth in the Netherlands (2.7%-points) is only partly due to the changed sector structure (0.5%-points).

3.6 Measurement problems

The literature also suggests that the growth slowdown is not a real phenomenon but is a consequence of larger measurement problems of the new economy. Measuring productivity faces classic problems, e.g. the correction for quality improvements in the healthcare sector. But the problems have been exacerbated by the digitalization of the economy.[9] For example, before we could take digital photos, the productivity of developing and printing photos could be calculated. Nowadays, we store and view photos on our phones. How should the productivity of this 'free' service be calculated? However, calculations by Syverson (2021) indicate that measurement problems are probably insufficient to explain the productivity slowdown.

Measurement problems are specifically mentioned in the valuation of the increasing intangible capital stock. Intangibles consist of, for example, R&D, software, human and organizational capital. According to Brynjolfsson et al. (2021), errors in measuring productivity effects of investments in intangibles can be described with a J-curve. At the beginning of the project, productivity is underestimated because inputs are also used for the production of intangibles that are poorly measured. In the course of the project, productivity is overestimated because intangibles are also used for the production of measured tangibles. In the Netherlands, too, intangible investments now exceed tangible investments, making it more difficult to estimate TFP growth.

3.7 Concluding remarks

We conclude that the literature is still very uncertain about the true causes of the slowdown in productivity growth and about the development of trends in the future. It is still unclear what the contribution of each cause is to explaining the slowdown (see the approach in Goldin et al., 2024). And what is the role of links between the causes? The growing importance of intangibles seems to be related to the increasing market power of superstars and the reduced business dynamics. A second question is how permanent are the outlined trend developments? Will the technology pessimists or the optimists be right about the breakthrough of new, substantial innovations? Is the weakening of business dynamics a temporary

[9] See the recent overview by Martin et al. (2024) and also De Bondt et al. (2021, section 6).

or a long-term phenomenon? Given these uncertainties, CPB, as the NPB, will continue to study the low productivity growth.

4 Productivity differences across Dutch firms are large and persistent

Michael Polder (Statistics Netherlands)

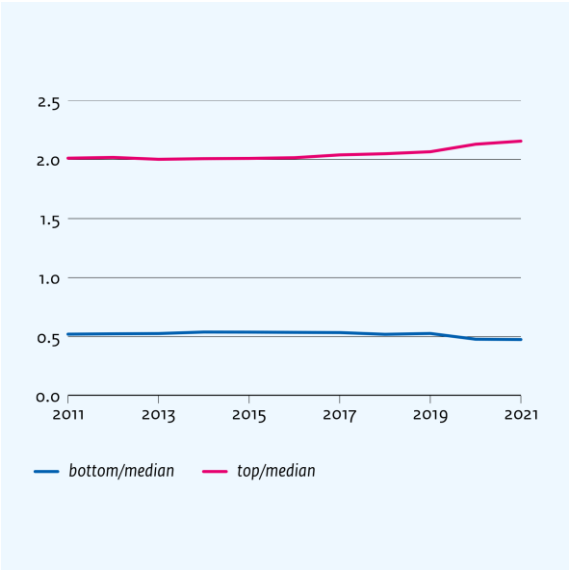
We present an update and extension of evidence on the dynamics of firm-level heterogeneity in labour productivity, and the relation between business dynamics and aggregate (industry-level) productivity. Understanding more about heterogeneity and the role of dynamics is essential for understanding aggregate growth. In particular, statistics about the within-industry dispersion in productivity are informative about whether the performance of firms in a particular industry is similar, or whether there are "winners" and "losers". A stylized fact in the literature is that the latter is the case (e.g. Syverson, 2011): the degree of heterogeneity in productivity is persuasive, even within narrowly defined industries . This is indicative for firm-level differences in things such as production technologies and innovation, as well as market frictions (Bartelsman and Wolf, 2018). In addition, productivity differences are found to be closely linked to differences in renumeration of workers (Cunningham et al. 2023). Finally, macroeconomic growth depends in part on creative destruction, through which the least productive firms are driven out of the market, making way for new innovative and more productive firms, as well as reallocation of market shares towards more productive incumbent firms (Aghion et al. 2021).

Note on data used
We use an experimental dataset on firm-level productivity that is currently being developed at Statistics Netherlands. This dataset is an updated and improved version of that used in the previous edition of the National Productivity Monitor (CPB, 2024). The outcomes presented should be regarded as experimental. In addition, due to differences in delineation of the research population, methodological choices (e.g. outlier exclusion), and variable definitions, these figures may not fully align with official statistics. The dataset currently covers the period 2011-2021 and concerns incorporated enterprises only (legal forms "bv" and "nv") in the business economy (excluding the financial and real estate sectors as well as petroleum). In the indicators we only consider firms with 3 persons (full-time) employed or more.

A decade of productivity dispersion

Frontier firms in the Netherlands are about twice as productive as the median firm, whereas the bottom group is about half as productive as the median. Productivity is measured as value added per hour worked. Dispersion is measured by differences between percentiles of the productivity distribution. In particular, we express in figure 4.1 the labour productivity level of the top decile and the bottom decile relative to the median level. It shows large differences between firms at the top and the bottom of the distribution.

Figure 4.1 Labour productivity, frontier (top-10%) and laggard (bottom-10%) firms compared to median, business economy, 2011-2021



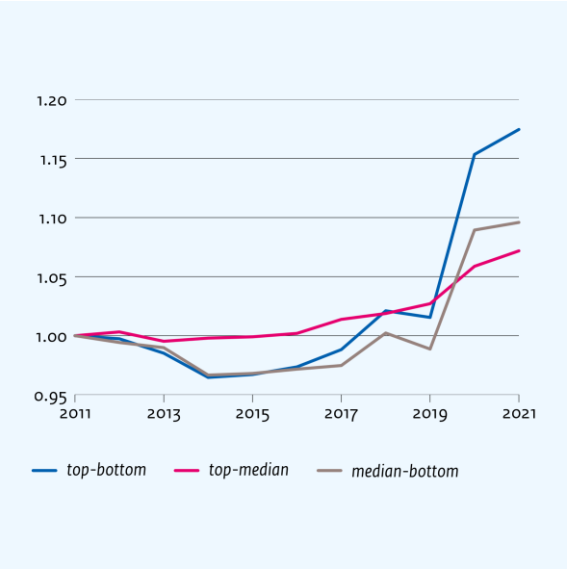
Note: top, median and bottom refer to respectively 90th percentile, median, and 10th percentile of the productivity distribution. Productivity is measured as real value added over persons employed in full-time equivalents. Percentile deviations are calculated within-industry (NACE Rev. 2, two digit) and year. Only enterprises with 3 or more persons employed and legal personality (corporations) have been included, that are part of the Business Economy (NACE Rev. 2 B-N, excl. K, incl. 95; we also exclude 19 and L). Nominal value added is deflated using National Accounts industry-level deflators. For more on the underlying data, see CBS (2022a, Chapter 2).

Growing dispersion in the Netherlands can already be detected before the Covid period. Figure 4.2 shows the change in dispersion relative to the base year 2011. While the figure confirms the stable pattern found in van Heuvelen et al. (2018) for up to 2014/2015, an increase in dispersion can be detected for more recent years. This trend started already before Covid. From 2015 to 2019, the top-bottom gap increased by on average 1.2% annually, composed of a 0.7% increase of the top-median gap and a 0.5% between median and bottom. Growing dispersion has been documented globally (Andrew et al., 2016), but there is variation across countries (Berlingieri et al., 2017) and over time (e.g. in the case of US manufacturing, Cunningham et al., 2023).

The dispersion increased strongly in 2020 and more gradually in 2021. When the economy was hit by the Covid-shock for the first time, this resulted in a substantial decrease in macroeconomic productivity (see figure 2.1).[10] Nevertheless, figure 4.2 shows that this impact has been uneven across the business population, resulting in a productivity spread much higher than in previous years and notably a higher gap for the lowest part of the distribution. Moving further into the Covid period, the dispersion continued to increase. The top-bottom gap grew in 2021 with 1.8%, mainly due to the frontier firms moving away from the median (1.2%).

[10] We note that employment is in full-time equivalents and refers to hours worked according to contract.

Figure 4.2 Labour productivity dispersion, business economy, 2011-2021 (2011 = 1)



Note: Ratios are expressed relative to the value in 2011.

Multinational firms are strongly represented in the frontier group. While the lion share of firms in the frontier group are independent small and medium enterprises (SMEs), their share is substantially lower than in the overall firm population (74.5% of frontier firms versus 83.6% in the population in 2021; table 4.1). Especially foreign multinational enterprises (MNEs) form a relatively large part of the frontier, compared to their population share (16.1% versus 9.3%). The post-2015 increase in dispersion has gone hand in hand with a slight shift within the frontier group, away from Dutch multinationals towards the other groups of firms.

Table 4.1 Composition of the frontier group.

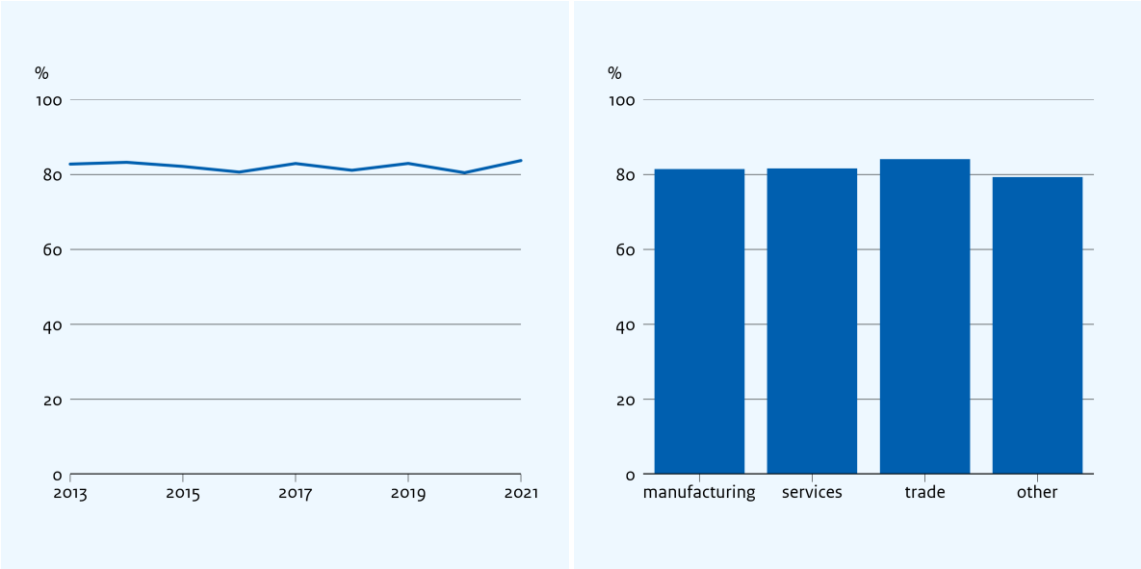
	independent SME	domestic large	Dutch MNE	Foreign MNE
Population				
2015	83.8%	0.5%	7.5%	8.2%
2019	83.6%	0.7%	7.0%	8.8%
2021	83.6%	0.7%	6.4%	9.3%
Frontier				
2015	73.1%	0.4%	10.8%	15.7%
2019	73.5%	0.5%	9.7%	16.3%
2021	74.5%	0.6%	8.7%	16.1%

Note: independent SME not part of a domestic group with 250 persons employed or more, nor part of a multinational; domestic large is a firm with more 250 persons employed or more and not a multinational; Dutch MNE is an enterprise under Dutch ownership (head office and/or affiliates); foreign MNE is an affiliate of a foreign multinational.

We find that persistence at the frontier is substantial. The composition of the frontier group changes each year. The results in figure 4.3 suggest that persistence is rather high with over 80% of firms remaining at the frontier from year to year, or reappearing after a single year.[11] This rate has been rather stable over time and across sectors.

[11] These numbers are lower than in the previous edition due to improvements in the data. It should be noted that we now also include firms reappearing at the frontier after one year as part of persistence. However, also without these additional cases the rates remain

Figure 4.3 Persistence at the productivity frontier (fraction of firms that stays in top decile)



Note: For these calculations, labour productivity was smoothed by taking the average over two consecutive years. When firms return to the frontier after a single year, this has also been counted as persistence. Figure by sector concerns averages across the period 2013-2021. NACE codes manufacturing: 10-33, ex. 19; services: 49-82 + 95, ex. K and L; trade: 45-47; other: 35-43.

Industry productivity growth: the contribution of business dynamics

The productivity growth of industries is composed of the contribution of continuing firms, new firms and closing firms. Figure 4.4 shows a decomposition of sectoral labour productivity growth for the period 2011 to 2021. The contribution of continuing firms is broken down further into a within component (i.e., the contribution of productivity growth within those firms), and a between component (i.e., the gain in aggregate productivity due to changes in the relative size of firms). The contributions of entry and exit are broken down further into those pertaining to actual firm birth or deaths, and those that are due to other reasons (such as M&A, restructuring and split-offs).

There are sectoral differences in the contribution of business dynamics. In manufacturing and in industries grouped under "other" (Construction; Energy; Water; Waste), productivity growth was mainly driven by within-firm productivity growth. In trade, there is an important role for exit of below-average productivity firms. In services, there is a relatively large contribution of newly founded firms achieving above-average productivity levels.[12]

We observe that the role of business dynamics for productivity growth declined in manufacturing and services. We define the contribution of business dynamics as the combined (net) contribution of the entry, exit and between components. Comparing the periods 2011-2015 to 2015-2019 in figure 4.4, we find a decline in the importance of business dynamics in manufacturing (from 3.2 to 1.7 %-point) and services (from 2.4 to 1.0%-point). This corroborates evidence for the US on declining dynamism discussed by Akcigit and Ates (2021) and Decker et al. (2017). However, we note that in services the lower contribution of business dynamics results

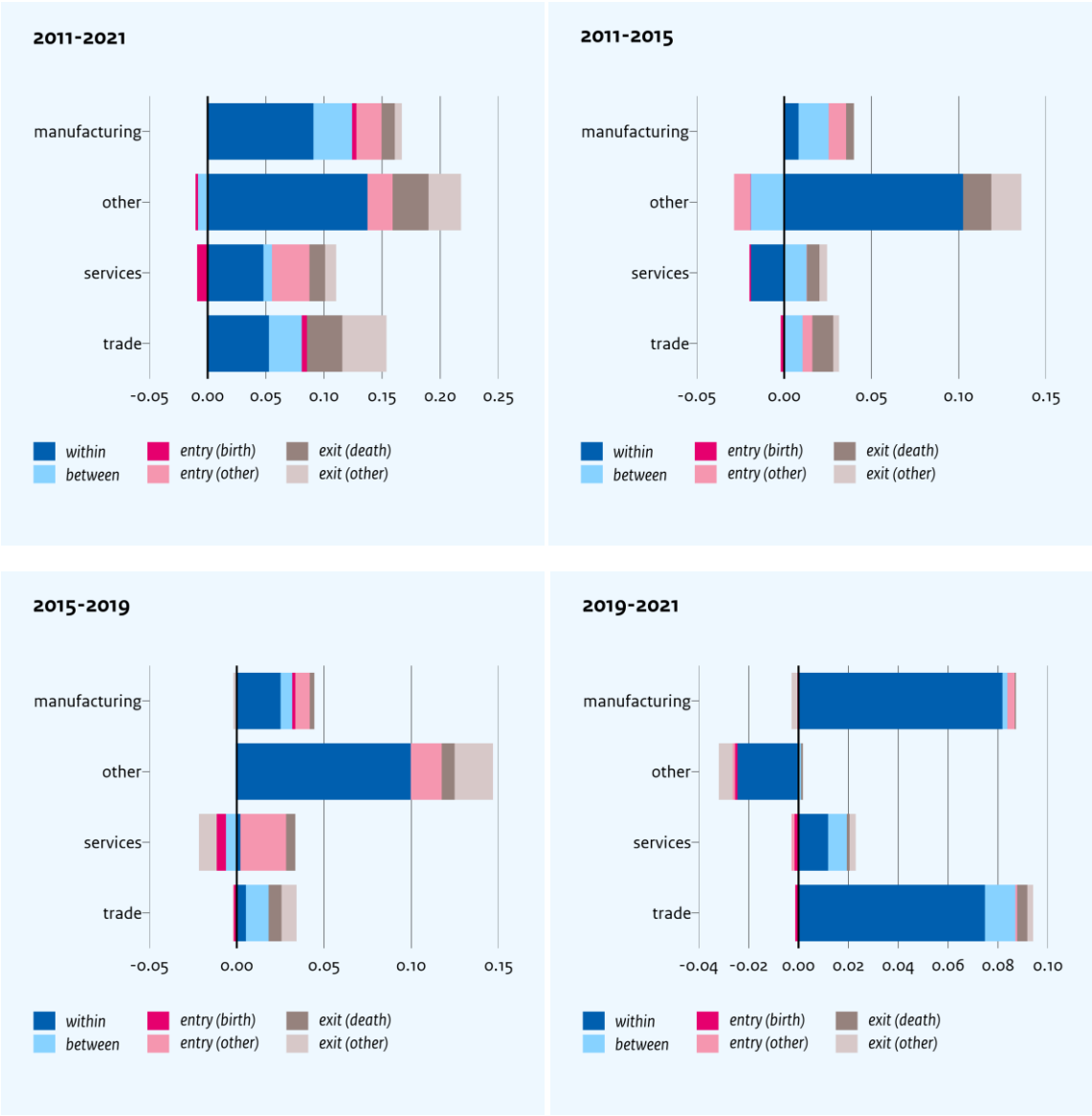
around 80%. With respect to the difference with results in Van Heuvelen et al. (2018) more research is needed. A candidate explanation is that unlike the earlier research, smaller firms (with 2 persons employed and less) have now been excluded. This is a particularly large and highly dynamic group, both in terms of business dynamics and productivity, which dominates the frontier group, and is likely to dominate the degree of dynamics observed as well. Moreover, we have improved the method to follow firms over time.

[12] Of course, considering a period of several years, there is more entry and exit than when looking at annual changes. Therefore, we find that the contributions of entry and exit are relatively large, compared to decompositions of annual changes or to contributions averaged across years.

from a mix of contributions, which does not necessarily indicate less dynamics. In the "other" industries, on the other hand, business dynamics became more important, while in trade it did not change much.

During the Covid crisis business dynamics contributed very little to productivity growth. Nearly all of the productivity growth in the period 2019-2021 was realized through within-firm changes in all sectors. Only in the group of "other" industries this resulted in a productivity decline. The contribution of exit is low. This finding is consistent with existing evidence on low productivity firms having had a higher likelihood of support during the Covid crisis (see e.g. Bettendorf et al., 2021).

Figure 4.4 Decomposition of labour productivity growth by sector



Note: Productivity is measured as real value added over persons employed in full-time equivalents. Percentile deviations are calculated within-industry (NACE Rev. 2, two digit) and year. Only firms with 3 or more persons employed and legal personality have been included, that are part of the Business Economy (NACE Rev. 2 B-N, excl. K, incl. 95; we also exclude 19 and L). Nominal value added is deflated using National Accounts industry-level deflators For more on the underlying data, see CBS (2022a; Chapter 2). Firms in the top and bottom percentile in either the productivity level or growth distributions have been excluded. The decomposition follows the method suggested by Foster et al. (2001).

5 Productivity benefits of export networks

Daan Freeman, Mark van der Plaat and Benjamin Wache (CPB)

5.1 Introduction

International trade remains crucial for the Netherlands. While concepts like strategic autonomy and reshoring have become an integral part of the debate on international trade, the export sector continues to play a vital role in the Dutch economy. Exporting firms are larger and more productive than the average Dutch firm and offer higher wages (CBS, 2022a). According to Statistics Netherlands (CBS), over 30% of gdp is generated through exports (CBS, 2022b). This added value stems not only from the exporting firms themselves but also from their suppliers. Approximately half of the jobs linked to exports are found within these supplier firms (CBS, 2022b). So, we know from previous research that exporting firms are more productive than non-exporting firms. This study extends the analysis to the whole export network.

We examine the productivity of exporters and the firms connected to them through supply networks. By focusing on firms connected to exporters, we gain insights into the broader impact of exports on the Dutch economy. Even firms indirectly involved in international trade may benefit through their connections with exporters. We emphasize productivity because it is a key driver of economic growth and competitiveness for firms and the Netherlands as a whole.

We use a new dataset from CBS that maps the Dutch production network, detailing domestic trade relationships between firms. The highly granular nature of this data allows us to uncover patterns at the firm level, rather than the traditional industry level. This enables us to identify which firms act as suppliers or customers of exporters and which do not. Suppliers to exporters, like exporters themselves, tend to be larger, more productive, and have more customers than firms not directly linked to exporters. This suggests benefits that extend beyond the (additional) revenue from supplying to exporters.

The structure of this section is as follows. Section 3.2 outlines the data sources underpinning our findings. Section 3.3 presents descriptive statistics, followed by results from several regression analyses. Section 3.4 places the findings in an international context and discusses potential explanations. The final section concludes with a summary and briefly highlights areas for future research.

5.2 Data

To map the connections between firms, we use a new dataset from CBS, the *Production Network of Businesses* (PNB). This dataset includes intermediate deliveries of goods and services between firms in the Netherlands, showing which firms trade with each other and the types of products exchanged. Firms combine various goods and services in their production processes, meaning suppliers can come from different industries. For instance, a machinery manufacturer might integrate metal components and construction advice from suppliers across different industries to develop and produce machinery. We focus on the flow of deliveries between firms in this network. This dataset is experimental. While a significant portion of this data

is based on direct observations of transactions between firms, a part is estimated using methods capable of reconstructing networks (Mungo et al., 2024). For a detailed explanation, see Buiten et al. (2021).

The PNB provides an unprecedentedly detailed view of Dutch value chains. Value chains are chains of production stages from raw materials to finished products. In existing literature, both national and international, chains are typically analysed at the industry level, describing for example the amount of agricultural inputs used in the food manufacturing industry (Buiten et al., 2023). CPB has extensively used such data in its research (e.g., van der Wal et al., 2024; Wache et al., 2024). The PNB takes this a step further by focusing on deliveries between individual firms, enabling the identification of differences within industries. This granular approach also allows for the study of dynamics such as spillover effects at the firm level.

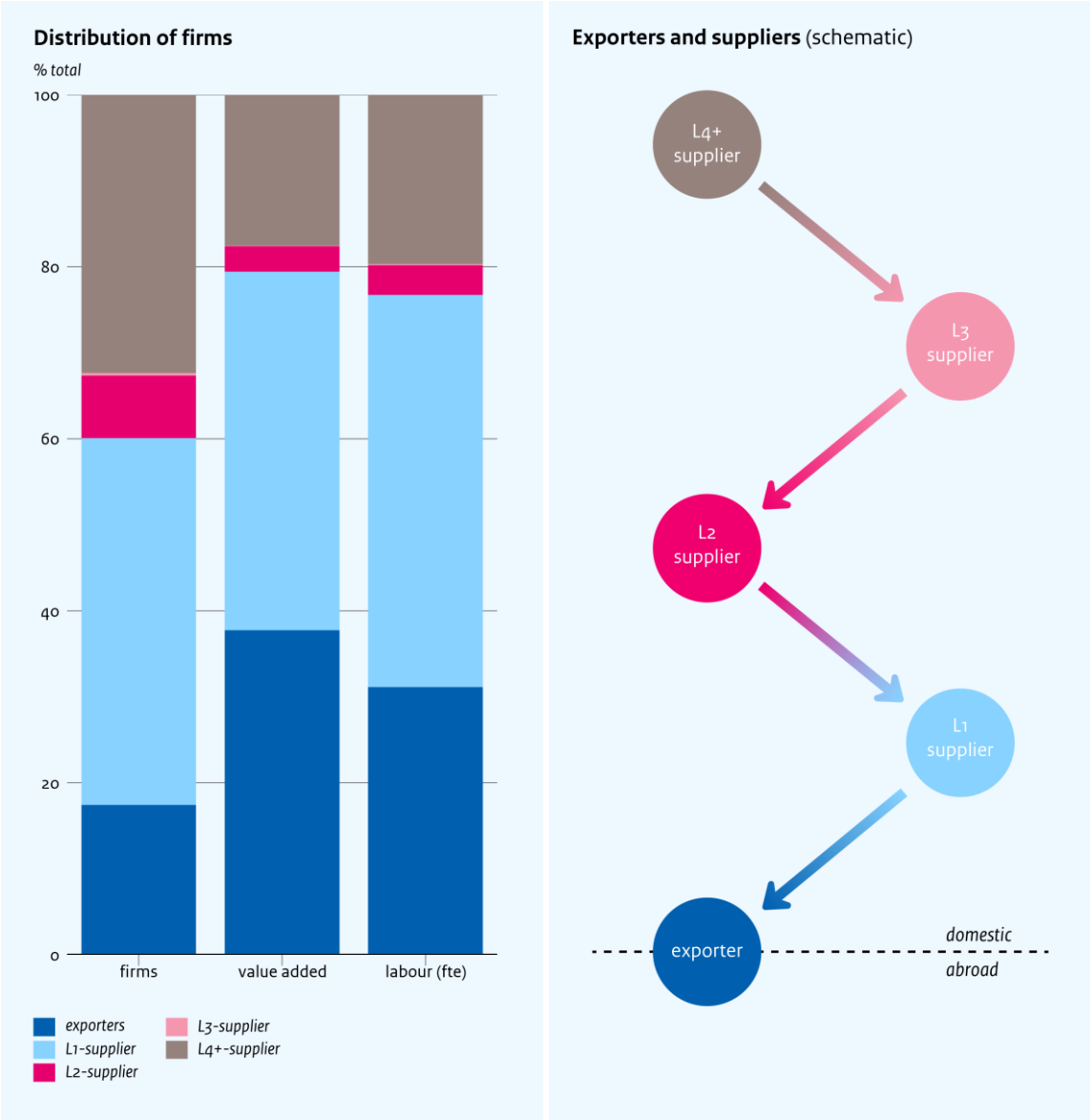
In addition to the network data, we use CBS data on firm-level productivity and international trade in goods. The productivity data includes labour productivity, total factor productivity (TFP), labour, capital, revenue, and value added. The dataset consists of data on corporations, thus excluding self-employed individuals and other non-corporate entities. Based on trade data, we identify exporting firms as explained below.[13] Our analysis is limited to goods exports, as data on international trade in services is unavailable.

The combined dataset includes around 55,000 firms for the analysis year 2018. A firm is classified as an exporter if it was actively exporting goods for at least three years between 2015 and 2018, to exclude one-time exporters.[14] Exporters in the trade and transport industries are excluded, as these often act as intermediaries. Additionally, firms with fewer than five employees are excluded. These criteria result in a higher proportion of exporters (17%) in the dataset compared to the overall firm population (10%) (CBS, 2022b).

Importantly, we identify not only direct suppliers to exporters but also firms supplying these suppliers and beyond. Around 17% of firms in our data set are exporters. So-called L1 suppliers, which supply directly to exporters without exporting themselves, account for over 40% of the total (see Figure 5.1). L2 suppliers, those supplying to L1 firms (but not directly to exporters or exporting themselves) represent a smaller group (7%). Similarly, we identify L3 suppliers, which form an even smaller group (less than 1%). Firms in the L4+ group, which are minimally connected to exporters, comprise about 30% of the total firm population. Together, exporters and L1 suppliers account for around 60% of total number of firms, 80% of the total added value and employment (Figure 5.1).

[13] Re-exports are excluded from our analysis. Firms solely engaged in re-export activities are therefore not classified as exporters.
[14] In the literature, such firms are referred to as *perennial exporters* (see van den Berg et al., 2022).

Figure 5.1 Exporters and their L1 suppliers constitute 60% of the firm population, 2018

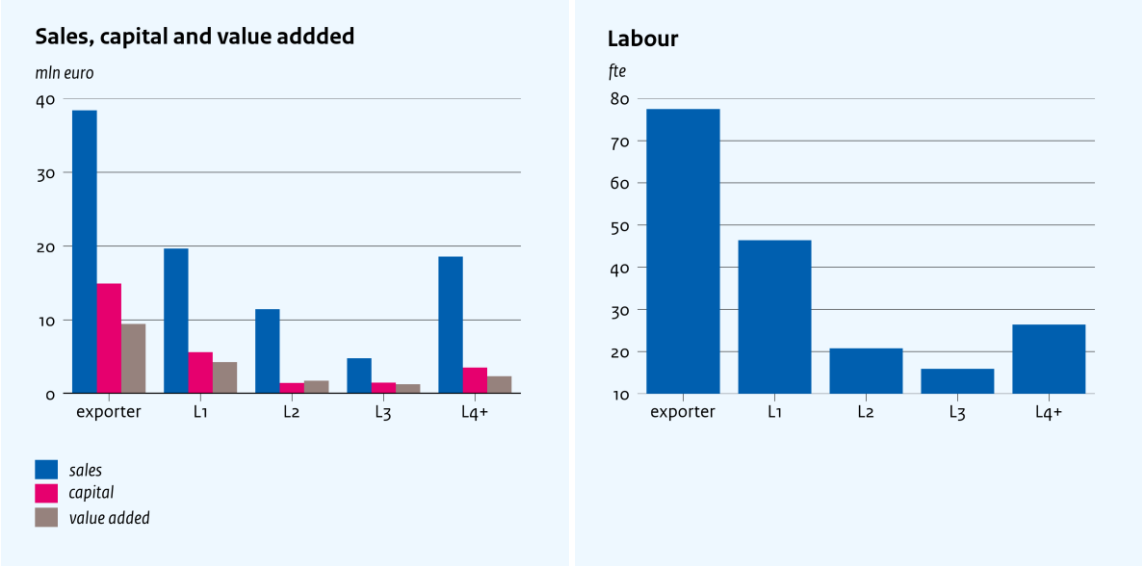


5.3 Results

Exporters are typically larger and more productive than non-exporters. They employ more workers, generate higher revenue, add more value, and maintain larger stocks of capital goods (see Figure 5.2). On average, exporters employ two to three times as many workers as non-exporters, enabling them to generate three to four times more revenue and value added. This higher value added per employee reflects their higher productivity. These findings align with existing literature, such as Bernard and Jensen (1999), which consistently shows that exporters globally tend to have more employees, larger capital stocks, and higher productivity than non-exporting firms. This higher than average productivity level is not so surprising given that exporting to markets beyond your domestic home market typically entails overcoming significant fixed costs due to international trade hurdles such as trade and transport costs, but also information, search and advertising costs.

Direct suppliers to exporters (L1) are also generally larger and more productive than firms that do not supply directly to exporters, although they are still smaller and less productive than exporters themselves. Our new finding is that suppliers to exporters are also larger and more productive than the average firm. L1-suppliers, on average, employ twice as many workers, generate double the revenue, and create two to four times more added value than firms not directly connected to exporters (see Figure 5.2). Their capital stocks are also two to three times larger. However, exporters outperform L1 suppliers across all these dimensions. Together, exporters and L1 suppliers form the most productive group of firms in the Netherlands. Firms further removed from exporters—such as L2 and L3 suppliers—are smaller and less productive than both exporters and L1 suppliers. The L4+ group includes a mix of firms, some of which are large retailers and wholesalers. These larger firms raise the group's average figures for employees and revenue, even though many other L4+ firms are smaller and less productive.

Figure 5.2 Exporters are larger and have more capital than other firms, 2018



Note: Average values per group.

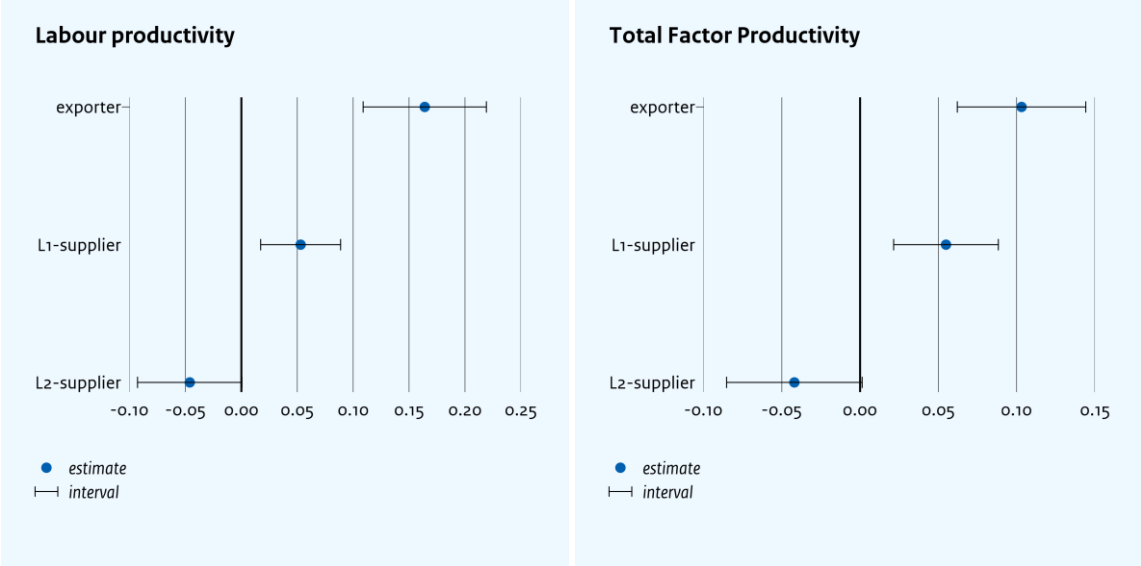
We perform several regression analyses to further examine differences between firms. The above figures compare firms of varying sizes and industry . Regression analysis allows us to control for the number of employees and industry fixed effects, creating a better comparison between groups. This approach ensures that comparisons between supplier groups are not biased by one group having more large firms or firms from specific industries.

Regression analysis enables the comparison of different groups of firms. The coefficients in Figure 5.3 represent the level difference relative to the L3+ supplier group. Given the small number of L3 suppliers, we combine them with the L4+ suppliers into a single group, together called L3+ group. So if we analyse labour productivity and find that the coefficient for exporter status is positive and statistically significant, this indicates that exporters, on average, have higher labour productivity than the L3+ group. The coefficient of each group reflects the percentage difference with the L3+ group. We stress that this analysis does not establish causal relationships.

After adjusting for various firm characteristics, exporters remain the most productive group, followed by their direct suppliers. Figure 5.3 presents the regression results for both labour productivity and TFP, with exporters showing the highest coefficients. On average, exporters are 11% to 18% more productive than the L3+ group. Direct suppliers (L1) are also more productive than other firms, by around 5% on average. Indirect

suppliers (L2), however, do not exhibit higher productivity than other firms. A clear pattern emerges: exporters are the most productive, followed by their suppliers, and then the rest.

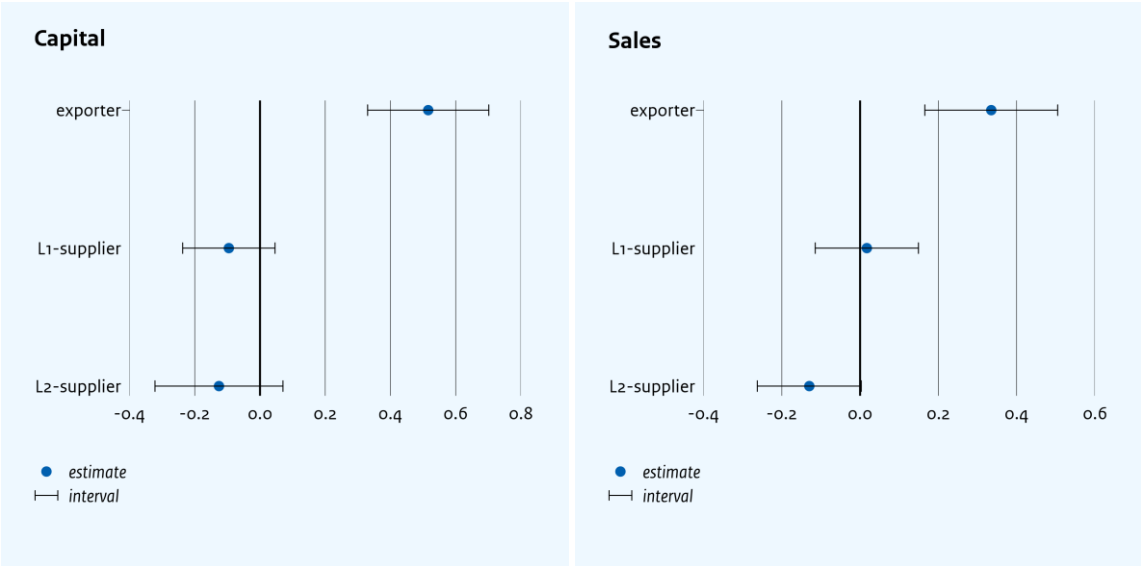
Figure 5.3 Exporters and their suppliers have higher productivity



Note: the whiskers show the 95% confidence interval

Exporters are not only more productive but also generate higher revenue and utilize more capital goods. Figure 5.4 presents the regression results for both sales and capital intensity. Compared to productivity, the patterns for sales and capital differ: exporters achieve the highest sales and have the largest capital stock. On average, exporters use nearly 70% more capital goods and generate 40% more sales than the L3+ group.[15] However, their direct (L1) and indirect (L2) suppliers do not have significantly more capital goods or higher sales. This indicates that exporters use their extensive capital stocks to achieve higher productivity than non-exporters.

Figure 5.4 Exporters are also larger and utilize more capital



Note: the whiskers show the 95% confidence interval

[15] A coefficient of 0.52 implies that capital is around 70% larger ([exp(0.52)-1]*100).

5.4 Explanations

The productivity hierarchy of firms in export networks is not unique to the Netherlands. For example, Dhyne and Rubínová (2016) show that exporters and their suppliers are larger and more productive than other firms in Belgium. There is also a clear hierarchy: exporters are larger and more productive than their suppliers, who, in turn, outperform their own suppliers. Similar trends are reported for Japan (Ito & Saito, 2021). At the industry level, Merlevede and Theodorakopoulos (2021) observe comparable patterns across a broader set of countries.

Selection, economies of scale, and learning effects may explain why exporters tend to be larger and more productive than other firms. Three main drivers of differences between exporters and non-exporters are suggested in the literature. First, according to the *selection effect*, exporters face intense international competition, meaning only highly productive firms can afford to export (Melitz, 2003). Firm size is also critical, as shown by Brakman et al. (2020), who emphasize that firm size significantly impacts the likelihood of starting to export. Second, exporters access larger markets, enabling them to achieve *economies of scale* and lower production costs (Krugman, 1980). Third, by interacting and collaborating with foreign partners, exporters *learn* new techniques or methods, leading to productivity gains (Atkin et al., 2017; CBS, 2022a; De Loecker, 2013). Thus, exporting not only attracts productive firms but also enhances their productivity over time.

Similar factors apply to suppliers of exporters. Productive exporters are likely to invest more in identifying and collaborating with the most productive suppliers, creating a mutually beneficial selection process (Bernard & Moxnes, 2018). Additionally, exporters' economies of scale may benefit their suppliers by increasing the demand for intermediate goods, which boosts suppliers' production efficiency. Lastly, suppliers may adopt new techniques and practices through their partnerships with productive exporters, further improving their own productivity. Empirical studies confirm that exporters often assist suppliers in delivering higher quality and encourage continuous improvement (Alfaro-Ureña et al., 2022).

The relationship between exporters and their suppliers is mutually beneficial. Exporters gain advantages from working with productive suppliers, as highlighted by Bernard et al. (2022) and Bernard and Moxnes (2018). Their model describes how marginal costs of an exporter depend on supplier pricing. Productive suppliers offer lower prices, which keeps costs in the supply chain low. As a result, firms with productive suppliers in their value chains are more competitive than those without such networks. Additionally, productive firms attract more customers and achieve higher sales per customer.

5.5 Conclusion

Exporters play a crucial role in the Dutch economy, extending beyond their function as employers and customers of domestic suppliers. They make a significant contribution in particular to productivity, not only through their own performance, but also possibly by enhancing performance of suppliers. Exporters are the most productive firms in the Netherlands, followed by their suppliers, who also perform above average. The findings suggest that the closer a firm is connected to the export sector, the more it can benefit. Firms gain from exports through various channels, each with distinct policy implications.

The insights from this study are relevant to several policy debates. Policies affecting exports and exporters have broader impacts, influencing not just the export sector but the wider economy. To boost productivity, the

government could, for instance, focus on promoting knowledge diffusion from successful exporters to other firms or facilitating stronger connections between firms. Additionally, stimulating the export sector itself can indirectly drive growth across the broader economy. Effective policy design requires a deeper understanding of productivity spillovers from exporters to their suppliers and beyond. Future research should explore these channels in detail.

Future research will delve further into the differences between exporters and their suppliers, with particular emphasis on the diversity within these groups. First, investigating how suppliers benefit from their relationships with exporters will help design more targeted pro-productivity policies. Second, identifying which firms derive the most advantage from working with exporters—considering factors like firm size, industry, export destinations, or specific products—will clarify the links between (indirect) exporting and productivity. Finally, examining non-export-connected firms will provide insights into their role in the economy. These insights will contribute to the development of more focused policies that support productivity and economic growth in targeted areas of the economy.

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We expect EUR bank bond supply to reach €450bn in 2027. Redemptions remain the main driver of the high supply, but the modest macroeconomic outlook is not expected to push supply up any further. We foresee covered bond issuance reaching €185bn, senior unsecured bond supply at €210bn and another €55bn in subordinated bank bonds to be printed

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Variables driving next year's bank bond supply

We expect EUR bank bond supply to reach €450bn in 2027, down slightly from the €455bn forecast for 2026. In our view, several variables will be steering next year's issuance.

Redemptions will remain the central driver of the elevated 2027 bank bond supply forecast. Nearly €400bn of bonds mature in 2027, up €30bn year-on-year and the highest level in a decade.

The increase is especially significant in the covered bond segment where €186bn will mature, a historic high.

The subordinated segment will also rise, with redemptions totalling €45bn, up about €6bn from 2026. Of this total, €31bn is in Tier 2 instruments and €14bn in AT1s. The senior unsecured segment will partially offset these increases, with total redemptions reaching more than €162bn, down from more than €170bn recorded this year. The majority stems from senior bail-in bonds (senior non-preferred or senior HoldCo) with more than €90bn and an additional €71bn maturing in senior preferred instruments.

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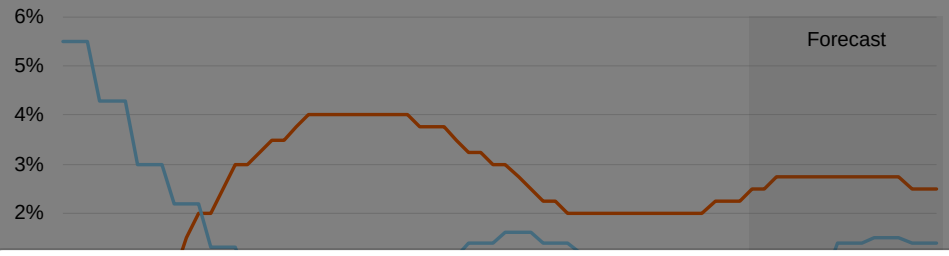
While high redemption levels point to stronger supply in 2027, other factors also play a role, such as the macroeconomic environment. Eurozone GDP growth is forecast to increase slightly in the second half of next year to reach 1.5%, still well below levels seen after the pandemic.

Furthermore, our economists pencil in an additional European Central Bank rate hike in 2026, bringing policy rates to 2.75% in early 2027. The higher interest rate environment is not expected to

support lending growth, especially as geopolitical instability remains. As a result, we do not see the modest macroeconomic backdrop acting as a catalyst for bank bond supply in 2027.

Policy rates and GDP growth forecasts



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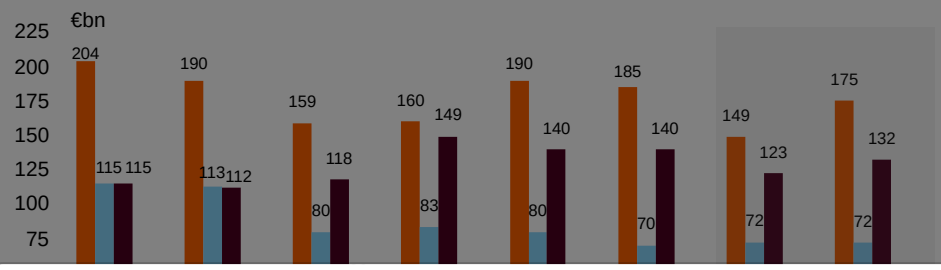
led to spreads widening again. The combination of French political uncertainty and the Middle East conflict could trigger some periods of volatility in the market in 2027. However, we believe these uncertainties will not be a strong negative for bank bond supply. Depending on the timing, though, periodic widening may make financial institutions less keen to pre-fund redemptions coming up in 2028.

Our 2027 supply forecast

When accounting for these three main variables, we expect the EUR bank bond supply to reach €450bn in 2027. This translates into a net supply of €57bn, down from €89bn in 2026.

Forecast EUR bank bond supply for 2027



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bond issuance. Besides, with the securitisation package entering its finalisation stage, issuers with high funding needs may shift some secured issuance towards securitisation for diversification purposes. Increasing demand for covered bonds from overseas investors driven by regulatory developments may also result in more currency diversification, away from EUR-denominated issuance.

We forecast senior unsecured supply to reach €210bn next year. This is split between €70bn in senior preferred bonds and €140bn in senior bail-in instruments. It represents a €10bn YoY decline in

senior preferred supply, reflecting both the drop in the segment's redemptions from 2027 onwards and the potential impact of the Crisis Management and Deposit Insurance (CMDI) framework. As the CMDI reforms are expected to enter into force in 2028, they could reduce the attractiveness of senior preferred bonds versus senior bail-in instruments in some EU countries, as senior preferred exposures will rank lower than, and no longer pari passu to, uncovered deposits in the creditor hierarchy. The funding cost advantage of senior preferred unsecured issuance could decline. At the same time, issuing senior non-preferred bonds could serve the purpose of increasing the bail-inable layer protecting preferred s

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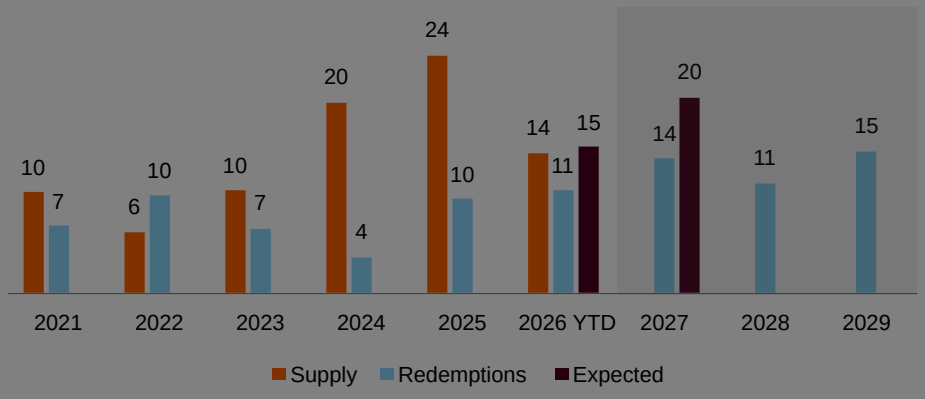
redemption levels. With AT1 redemptions in 2027 increasing by €3bn YoY to €14bn, net supply is likely to be marginally higher next year. In Tier 2, redemptions increase by €4bn to €31bn, essentially meaning that net supply will only be marginally higher.

As the chart below highlights, net supply in AT1 has been significant in recent years but has slowed this year as the focus has shifted towards refinancing. As such, while we expect more AT1 supply next year compared to 2026, the levels are relatively in line with the longer-term trend.

Forecast EUR AT1 supply (€bn)



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complementary role in capital management, and we do not expect it to have a major impact on bank capital issuance in 2027.

Bank M&A could also be a wildcard for 2027 issuance. With several large M&A transactions potentially taking place, this may lead to some capital optimisation and less incremental supply needs; conversely, goodwill effects, potential regulatory increases in buffer requirements or RWA inflation may lead to a slight increase in the need for capital. However, on the whole, we would expect M&A to have a slight dampening effect on supply.

Finally, spreads are currently quite elevated, and coupons on new issues are generally higher than legacy bonds maturing or callable in 2027. This may also lead to a more wait-and-see approach from issuers, but we still think that most issuers will continue to drive their capital plans as in previous years.

Forecast EUR bank bond supply for 2026

	FY 22	FY 23	FY 24	FY 25	FY 26F	FY27F	2025 YTD	2026 YTD
Covered bonds	204	190	159	160	190	185	149	175
Covered redemptions	*143*	*126*	*117*	*134*	*157*	*186*		
Bank senior unsecured bonds	230	225	198	233	220	210	195	204

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European TMT gross bond supply may reach €150bn in 2027, driven by tech funding for AI and data centres rather than traditional telecoms. Strong cash flows and credit quality support issuance, but disciplined funding will be key to market absorption

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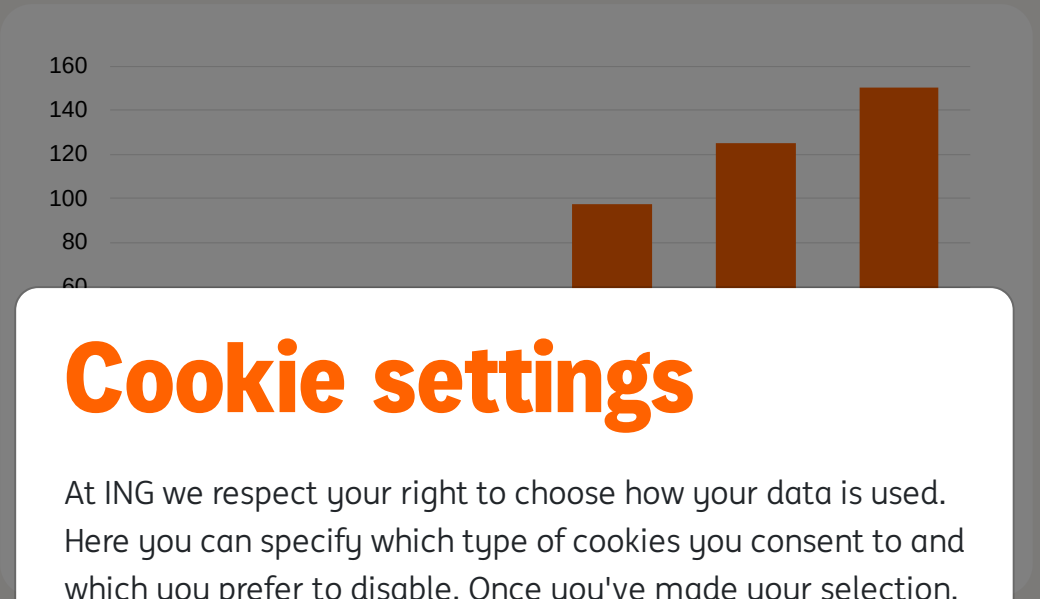
We predict another year of strong supply for the European TMT sector. We expect €150bn of gross issuance and €103bn of net issuance, given €47bn of debt redemptions. The additional issuance is not coming from the traditional telecom or media issuers, but from global tech companies.

As we expect the data centre build-out to continue in 2027, issuers are likely to return to the European investment-grade credit market to raise part of the required funding. Meanwhile, we do not see a major reason for traditional sectors to reduce or increase

indebtedness, as factors driving indebtedness higher are countered by factors driving debt lower, as we'll explain below.

European investment-grade TMT supply (€bn)



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in FY27.

Low-single-digit EBITDA growth gives companies some capacity to issue additional debt while maintaining stable leverage. However, higher interest rates provide an incentive to keep debt levels in check. We therefore expect the increase in debt issuance to be driven mainly by M&A. Several debt-funded acquisitions of smaller networks could generate around €5bn of net issuance. A further €3bn could come from US operators raising euro-denominated funding for spectrum payments in 2027.

Technology supply

In 2027, we believe the AI build-out will continue. Tech companies currently report healthy returns on investment, demand for AI services continues to outpace supply, and companies want to maintain their market positions. We therefore expect further capex growth in 2027, with the large hyperscalers investing close to US$1tr in fiscal 2027.

As in 2026, we expect these companies to finance their investments largely from operating cash flow, but also through debt issuance, with additional funding coming from European

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on future funding plans, as spreads may only tighten once the market no longer expects substantial new supply. Nevertheless, current spreads look attractive relative to the issuers' high ratings, with much of the widening driven by supply and liquidity effects. Relatively wide CDS spreads also suggest that technical factors, rather than a material deterioration in credit risk, are weighing on valuations. This combination of high credit quality and attractive spreads should attract new money to the investment-grade market.

Challenge 1: Does debt issuance reduce credit quality?

First, we expect the hyperscaler data centre operators Amazon, Microsoft, Alphabet, Meta and Oracle to grow their combined EBITDA by US$230bn, or 29%, in FY27, to around US$1tr. Moreover, with the exception of Oracle, these companies are expected to operate with limited net debt, with Microsoft holding more cash than debt. The main caveat is their significant lease commitments, many of which are not yet reflected in balance-sheet lease liabilities because the underlying leases have not commenced. Given our aggregate EBITDA projection of around US$1tr for fiscal

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accommodate more supply?

Do European credit markets have the capacity to absorb additional supply from technology companies? We think the current share of technology issuers remains modest. Alphabet accounts for only 1.16% of the iBoxx € Non-Financials index, while Amazon and IBM account for 0.68% and 0.91%, respectively.

In the US, Oracle's share of more than 2% in the iBoxx $ Domestic Non-Financials index illustrates that individual technology issuers can reach materially higher benchmark weights, while several high-

yield indices apply a 3% issuer cap. In addition, many large technology companies have not yet raised funding from European credit markets.

We therefore see significant room for technology companies to increase issuance in European investment-grade markets. The inclusion of eligible new bonds in benchmarks also creates a strong technical incentive for benchmark-aware investors to participate in new deals, as not owning the bonds could increase their tracking error.

Benchmark sizes selected TMT names in E

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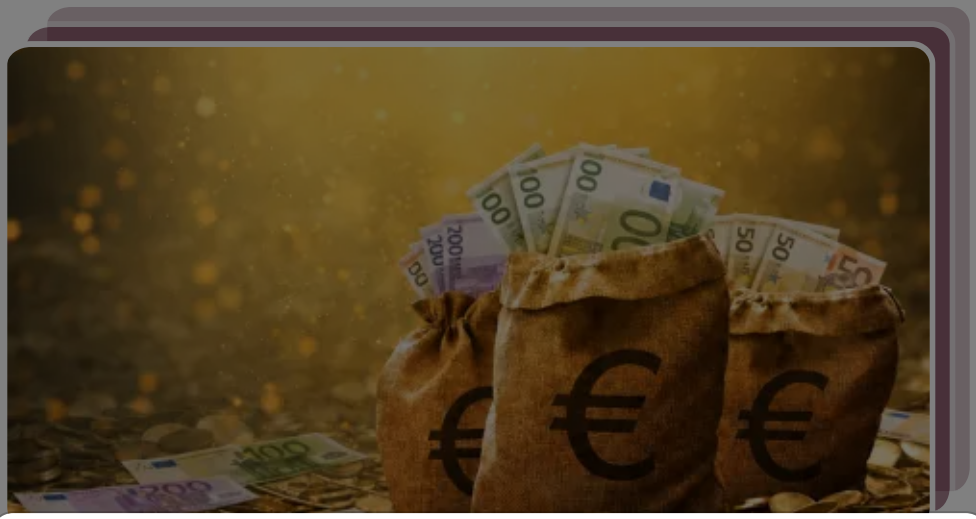

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